Lifelong Learning With Adaptive Knowledge Fusion and Class Margin Dynamic Adjustment for Hyperspectral Image Classification

Zihui Jiang, Zhaokui Li, *Member, IEEE*, Yan Wang, Wei Li, *Senior Member, IEEE*, Ke Wang, Jing Tian, Chuanyun Wang, and Qian Du, *Fellow, IEEE*

***Abstract*— With the rapid growth in satellite imagery acquisition and decreasing revisit intervals, efficient on-orbit processing of hyperspectral data has become critical due to limited onboard computing resources. In this context, lifelong learning (LLL) offers a promising solution to enable continuous learning from new data without storing all previous data or retraining from scratch. However, the plasticity–stability dilemma remains a significant challenge, particularly in hyperspectral image (HSI) classification under class-incremental scenarios. To address this, we propose a novel network architecture that integrates contrastive learning and an angular penalty loss. The contrastive learning module facilitates adaptive knowledge fusion, enabling the model to effectively incorporate new information while preserving prior knowledge. The angular penalty loss allows the classifier to dynamically expand for new classes while maintaining discrimination between old and new categories. Together, these components ensure robust knowledge retention, transfer, and adaptability. Experimental results on three benchmark hyperspectral datasets demonstrate that our method significantly outperforms existing approaches, highlighting its efficacy in addressing LLL challenges in HSI classification. The code is available at https://github.com/Li-ZK/LLL-AFCA.**

***Index Terms*— Contrast learning, hyperspectral image (HSI), knowledge fusion, lifelong learning (LLL).**

I. Introduction

HYPERSPECTRAL remote sensing captures both spatial and spectral information of ground objects, enabling the detection of subtle differences and making it a key focus in Earth observation technology. Classification is the core issue in hyperspectral image (HSI) processing and a key technique in its applications, such as agricultural analysis, natural disaster detection, military surveillance, and mineral exploration. [1], [2], [3], [4], [5]. Major classification methods, including maximum likelihood classifier, Bayesian classifier, decision trees, neural networks, and support vector machines, have been applied in HIS classification [6]. In recent years, with powerful feature extraction capabilities, deep learning models have successfully been applied in hyperspectral classification and become the mainstream in this field [7]. Deep learning models such as autoencoders, deep belief networks, convolutional neural networks (CNNs), recurrent neural networks, and generative adversarial networks are increasingly being applied in HSI classification [8], [9], [10].

With the improvement of remote sensing data collection capabilities, land-use and land-cover categories can be discerned, ensuring that classification models can quickly adapt to new data while retaining the discriminative ability for existing data. Additionally, as satellite revisit intervals shorten and data updating speeds increase, storing all previous data for further training poses a challenging task. Whenever a new category appears, training a new model on all existing and new data will greatly increase time and computational costs. In light of these circumstances, traditional classification paradigms [11], [12] are no longer effective.

Recently, lifelong learning (also known as incremental learning or continual learning) has been utilized to address the emergence of new classification tasks, learning efficient machine models from a series of gradually incoming data in training phases. Incremental learning enables the continuation of learning new class data based on knowledge of old categories. However, a drawback is that incremental learning models often tend to favor newly added data categories, leading to the forgetting of knowledge of previously learned old categories [13], known as the "catastrophic forgetting" issue. Xu et al. [14] have introduced knowledge extraction into HIS classification, allowing incremental knowledge learning of new classes without forgetting old ones. Additionally, a linear correction layer has been designed to balance relationships between different classes. Bai et al. [15] proposed an incremental learning method based on a geometric spatial–spectral feature integrated network and class-incremental learning framework, enabling the model to adapt to new class data without excessively forgetting knowledge of old classes

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Zihui Jiang, Zhaokui Li, Yan Wang, Ke Wang, and Jing Tian are with the School of Computer Science, Shenyang Aerospace University, Shenyang 110136, China (e-mail: jiangzihui@stu.sau.edu.cn; lzk@sau.edu.cn; wy4615@sau.edu.cn; wangke3@stu.sau.edu.cn; tianjing@stu.sau.edu.cn).

Wei Li is with the School of Information and Electronics and Beijing Key Laboratory of Fractional Signals and Systems, Beijing Institute of Technology, Beijing 100081, China (e-mail: liwei089@ieee.org).

Chuanyun Wang is with the College of Artificial Intelligence, Shenyang Aerospace University, Shenyang 110136, China (e-mail: wangcy0301@sau.edu.cn).

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: du@ece.msstate.edu).

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during training. Zhao et al. [16] introduced a lifelong learning strategy that continuously updates model weights using continuous spatial–spectral feature extraction. Yu et al. [17] presented a distillation-constrained prototype representation network for incremental classification, enhancing the discriminative ability of original classes while better adapting to new classes. Hou et al. [18] introduced the learning with unit cosine incremental rebalancing (LUCIR) unified classifier for incremental learning, which replaces the standard softmax layer with a cosine normalization layer to alleviate bias. Rebuffi et al. [19] proposed incremental classifier and representation learning (iCaRL) to select data subsets to retain by computing class averages. Liu et al. [20] introduced a novel network architecture called adaptive aggregation network to address the retention of old class knowledge and adaptation to new class data in incremental learning. Peng et al. [21] addressed the few-shot class-incremental learning task from an open-set perspective and achieved good clustering features by designing an angle penalty loss.

Although existing incremental learning methods have made some progress in alleviating catastrophic forgetting [1], they still face serious challenges in balancing the retention of old class knowledge and the adaptability to new class data. Class-incremental learning requires models to classify all classes in all learning tasks, thus class-incremental learning methods must maintain old class knowledge while learning new class knowledge. When the model focuses too much on learning new class knowledge, catastrophic forgetting can occur, leading to a sharp decline in classification performance on old classes. However, focusing too much on retaining old class knowledge can make the model less adaptable to new class data. Additionally, with more classes being present, the space reserved for new classes in the potential feature space will become increasingly limited, resulting in a decrease in the model's adaptability to new classes.

To address the above issues, we propose a novel lifelong learning method with adaptive knowledge fusion and class margin dynamic adjustment (LLL-AFCA) for HSI classification. First, we build an adaptive knowledge fusion network based on contrastive learning, which can simultaneously consider retaining old class knowledge and adapting to new class data. This network consists of multiple cascaded knowledge retention blocks, knowledge adaptation blocks, and knowledge fusion blocks. In the knowledge retention blocks, we freeze certain parameters to retain old class knowledge, while in the knowledge adaptation blocks, we update all parameters to learn new class knowledge, and the knowledge fusion blocks are used to integrate old class and new class knowledge. Specifically, since the lower level features of convolutional networks are often more universal, we introduce a contrastive learning block in the lower level network to learn general knowledge that can be adapted to both new and old classes. Furthermore, we design a dynamic angle penalty loss that dynamically adjusts class margin based on the number of categories, thereby reserving more space for new classes in the latent feature space.

The main contributions of the proposed LLL-AFCA are as follows.

1) A novel LLL-AFCA is proposed for HSI classification, which can alleviate catastrophic forgetting phenomenon.
2) We design an adaptive knowledge fusion feature extraction network based on contrastive learning, which can adapt to new class data while retaining old class knowledge.
3) We develop a dynamic angular penalty loss to dynamically adjust class margin, thereby creating more space for new classes in the latent feature space.

Extensive experiments are performed to demonstrate that the proposed method can continuously learn the additional classes' information and achieves superior classification results compared to current state-of-the-art CIL methods.

The rest of this article is organized as follows. Section II describes some recent research works. Section III provides a detailed explanation of the LLL-AFIA. Section IV showcases the experimental results on four publicly available HSI datasets. Finally, Section V concludes the article.

II. Related Work

A. HSI Classification

In recent years, due to their powerful feature extraction capabilities, deep learning models have achieved successful applications in the field of hyperspectral classification, becoming the mainstream method in this field [22]. Early applications included those using stacked autoencoders and deep belief networks. Chen et al. [23] proposed a deep architecture based on stacked autoencoders to combine low-level features. Zhong et al. [24] introduced the deep belief network for classification, which can extract more robust features and achieve classification accuracy superior to other models. Since both stacked autoencoders and deep belief networks employ fully connected layers with a large number of parameters, they require substantial training data. Moreover, these two deep models are not effective in processing spatial information. CNNs, which utilize local connections and weight sharing, can effectively extract spatial features and have gradually taken a major position in HSI feature extraction methods. Chen et al. [25] adopted a 3-D CNN to directly extract deep spectral–spatial features from raw HSIs, obtaining satisfactory classification results. Zhao and Du [26] proposed to use CNNs and dimensionality reduction methods for the extraction of spectral and spatial features from hyperspectral remote sensing images. Zhong et al. [27] proposed spectral and spatial residual blocks to continuously extract spectral–spatial features in HSIs, to some extent solving the problem of reduced classification accuracy due to network depth. However, this method extracts spatial features from the extracted spectral features, which may lead to the loss of spatial texture information in the images. Zhu et al. [28] proposed a residual spectral–spatial attention network for hyperspectral remote sensing image classification, achieving good classification results. Recently, Transformer models based on the multihead self-attention mechanism have attracted widespread attention from researchers. Hong et al. [29] used multiple adjacent bands as sequential inputs to the Transformer, capturing the long-distance dependencies of spectral features and confirming the effectiveness of Transformers in extracting these features.

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Tu et al. [30] introduced a neighborhood pixel aggregation module to address the challenge in extracting hyperspectral spatial features by providing a multiscale spatial neighborhood feature representation for the Transformer model.

B. Lifelong Learning

Early lifelong learning (LLL) methods [31] can be primarily categorized into three types: rehearsal, regularization, and parameter isolation approaches.

Rehearsal approaches involve storing data or features from the previous phase and replaying them during the new phase along with new data to prevent catastrophic forgetting. It is worth noting that the storage capacity for old data is limited. Bang et al. [32] introduced a novel memory management strategy called rainbow memory (RM), which continuously updates the retained old samples using current tasks and enhances the diversity of the old samples through data augmentation. Liu et al. [33] proposed a novel and automatic framework called mnemonics for multiclass incremental learning. They automatically select old data to be retained in an end-to-end manner and continuously update the retained old data during the new phase. Yu et al. [34] estimated the centers of each class using the semantic drift compensation (SDC) algorithm without storing images and performed classification predictions using the nearest class mean (NCM) method. Rehearsal approaches not only rely on the stored old data but also suffer from the imbalance between new and old class data, which biases the classification results toward new classes with larger amounts of data. This imbalance exacerbates catastrophic forgetting. Our method mitigates catastrophic forgetting more effectively by retaining a portion of the data and freezing part of the network, rather than solely relying on storing data or features.

Regularization approaches introduce an additional regularization term in the loss function to consolidate prior knowledge while learning new data. Kirkpatrick et al. [35] proposed elastic weight consolidation (EWC), which estimates importance for each parameter in the network to balance the importance of learning new and old knowledge. Li and Hoiem [36] introduced the learning without forgetting (LwF) method, which uses the output of the old model as soft labels for the old data and employs knowledge distillation to alleviate forgetting and knowledge transfer. Building upon LwF, Zhang et al. [37] proposed deep model consolidation (DMC) that applies a double distillation loss based on the asymmetry of the new and old data. Regularization approaches require a certain degree of correlation between new and old data to effectively apply old knowledge to new data. Our method, on the other hand, extracts common features of new and old classes in the shallow layers of the network through contrastive learning, while capturing distinctive features of new and old class data in the deep layers, eliminating the need to consider correlations between new and old data.

Parameter isolation approaches provide different model parameters for each phase of data to prevent forgetting. When there are no restrictions on the architecture size, new model branches can be added for new data while freezing the parameters of the old model [38]. Alternatively, the network structure can remain static, with fixed parts assigned to each task. Tao et al. [39] proposed the topology-preserving knowledge incrementer (TOPIC) framework, which utilizes topology structure for feature extraction and classification to learn relationships between different classes. Parameter isolation approaches often freeze a large number of parameters. In contrast, our method only freezes a portion of the network parameters to retain information, while allowing the rest to be continuously updated. This approach enables the retention of old knowledge and the learning of new knowledge simultaneously. Furthermore, during the incremental stage, there is no need to expand the network size.

The latest LLL method [40] summarizes a new approach, namely bias-correction approach, in addressing the catastrophic problem caused by class imbalance. This is primarily due to the fact that, at the end of training, the network encounters a significant amount of new data in a new phase, but it sees a small amount or no old data due to limitations such as memory. As a result, the classification results naturally tend to favor the new classes with larger data volumes. Hou et al. [18] introduced learning a unified classifier incrementally via rebalancing (LUCIR), which replaces the standard softmax layer with a cosine normalization layer to mitigate biases. Zhao et al. [41] proposed an incremental learning method based on a constantly updated classifier. Bias-correction approaches often train specific hyperparameters for each dataset. Our method adopts different data strategies to train the network. First, we train the main modules of the network using all accessible data. Then, the fusion parameters for new and old class knowledge are trained using an equal amount of new and old class data. This allows the network to adaptively learn knowledge.

III. Proposed Approach

To balance the retention of old knowledge and the adaptation to new knowledge, we propose an LLL method based on adaptive knowledge fusion and class margin dynamic adjustment. The detailed LLL-AFCA method is elaborated in this section. Specifically, it includes data augmentation, an adaptive knowledge fusion network based on contrastive learning (which includes knowledge adaptation blocks, knowledge retention blocks, contrastive learning blocks, and knowledge fusion blocks), and a dynamic angular penalty loss. In addition, a bilevel optimization program (BOP) based on different data strategies is proposed for model optimization.

A. Our Framework

The model training framework of LLL-AFCA is shown in Fig. 1. Assume the HSI input during the base phase$_0$ is D_0. First, data augmentation is performed on D_0, where two data are randomly selected, and new data and labels are constructed via interpolation. Then, the augmented data is input into a single residual network to extract features and compute class means. Using the NCM, Q data are selected from D_0 as representative instances and stored, denoted as ε_0. This step retains the primary knowledge learned from D_0. Finally, the extracted features are fed into a classifier, and the model is optimized using cross-entropy loss.

Assume the HSI input during the incremental phase$_i$ $(i \geq 1)$ is $D_i (i \geq 1)$ and $\varepsilon_{0,1,\ldots,i-1}$. These are input into the adaptive knowledge fusion network based on contrastive learning.

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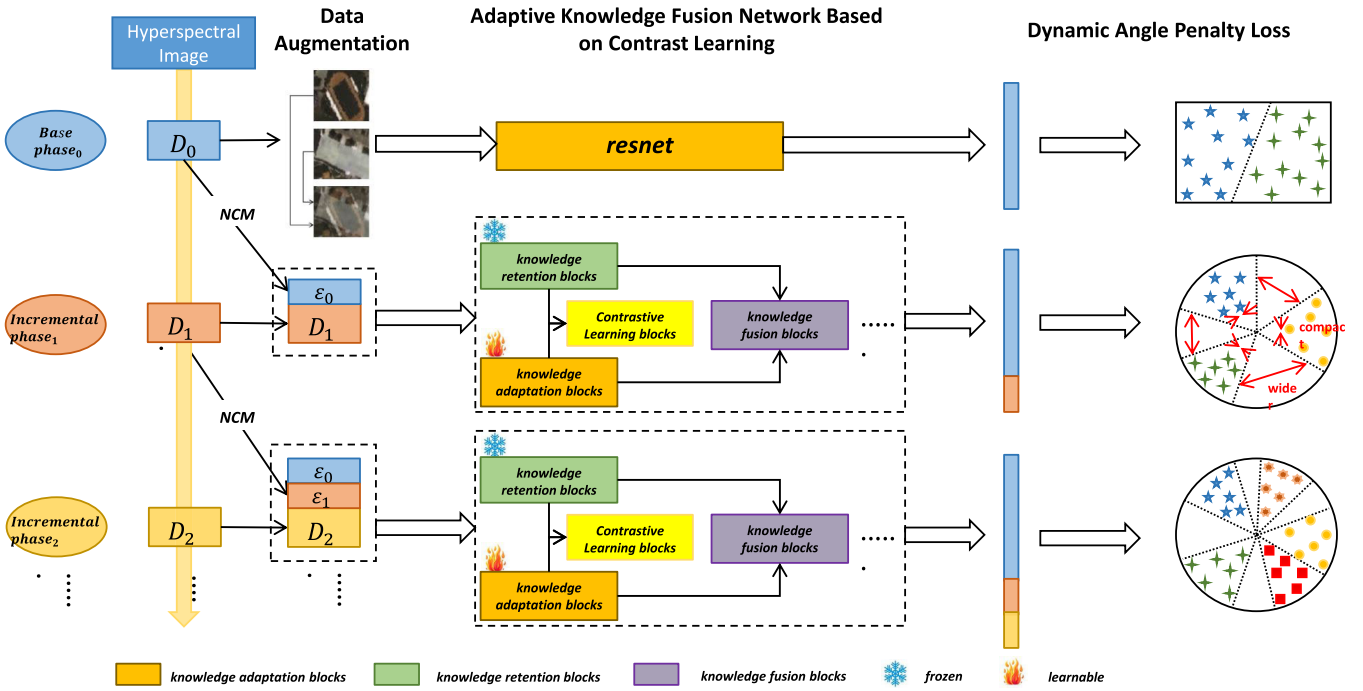


Fig. 1. Model training framework of LLL-AFCA. In the foundational stage, data augmentation is first applied, followed by feature extraction using a residual network. The model is optimized with cross-entropy loss, and class-nearest mean method is used to retain samples. In the incremental stage, the foundational network is frozen as a knowledge retention block to preserve old knowledge, while a duplicate network with the same architecture as the foundational network is used as a knowledge adaptation block to learn new knowledge. The network is divided into three levels. At the first level, a contrastive learning block is added, enabling the shallow layers of the network to learn shared knowledge between old and new classes. Finally, dynamic angular penalty loss is applied in the latent feature space to allocate more space for the new classes.

In the shallow network, the features learned by the knowledge adaptation block and the knowledge retention block are first fed into the contrastive learning block, and then into the knowledge fusion block, to learn the general knowledge of both old and new data. In the deep network, the features extracted by the knowledge retention block and the knowledge adaptation block are directly input into the knowledge fusion block to learn the distinct knowledge of old and new data. Finally, the class means are calculated from the extracted features, and Q data are selected from D_i using the NCM and stored as $\varepsilon_{0,1,\ldots,i}$. The model is then globally optimized using the dynamic angular penalty loss.

B. Data Augmentation

To improve the flexibility and generalization of the model, it is enabled to accurately classify both current and incremental data. We designed a data augmentation strategy using mixup to expose the model to a wide variety of classes during the base phase$_0$. Specifically, we randomly select two data, x_i and x_j, from D_0, and construct new data x_k through interpolation, which can be expressed as follows:

$$x_k = \lambda x_i + (1-\lambda)x_j \tag{1}$$

$$y_k = \lambda y_i + (1-\lambda)y_j \tag{2}$$

where x_i and x_j are two different training data, y_i and y_j are their corresponding labels, λ is a random number following a beta distribution, and x_k and y_k are the newly generated data and its corresponding label, respectively.

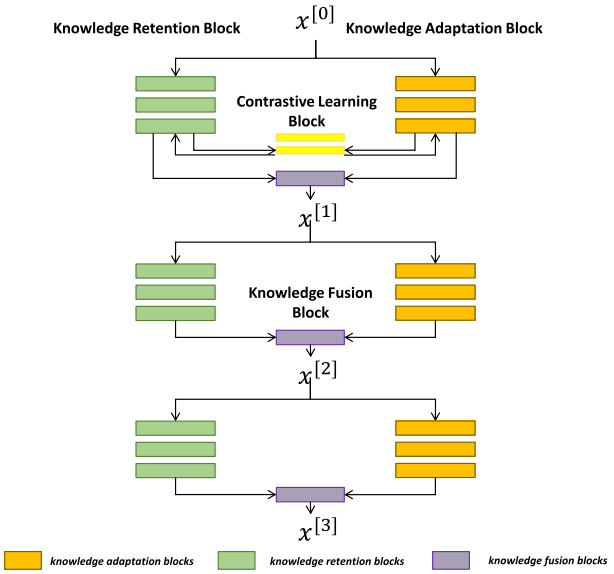


Fig. 2. Adaptive knowledge fusion network based on contrast learning.

C. Adaptive Knowledge Fusion Network Based on Contrast Learning

The feature extraction network framework for the incremental phase$_i$ is shown in Fig. 2. The adaptive knowledge fusion network based on contrastive learning consists of knowledge retention blocks, knowledge adaptation blocks, contrastive learning blocks, and knowledge fusion blocks. The knowledge retention block is derived by freezing and scaling the residual network from the base phase$_0$. The knowledge adaptation block is initialized from the residual network of

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the base phase$_i$. The contrastive learning block consists of fully connected layers. The knowledge fusion block functions as a fusion function for the knowledge fusion block and the knowledge adaptation block.

The network is divided into three levels. Let the input to the first level be denoted as $x^{[0]}$. $x^{[0]}$ is first fed into the knowledge retention block and the knowledge adaptation block separately to learn both old and new knowledge. The extracted features are then simultaneously input into the contrastive learning block, where general knowledge of both old and new data is learned in the shallow network. Next, the features are input into the knowledge fusion block, integrating old and new knowledge to produce $x^{[1]}$. Finally, $x^{[1]}$ is sequentially fed into the second and third levels to obtain $x^{[3]}$. It is worth noting that the second and third levels only contain the knowledge retention block, the knowledge adaptation block, and the knowledge fusion block to learn differentiated knowledge of old and new data.

1) Knowledge Retention Blocks: To retain old knowledge, the knowledge retention block is designed. In the incremental phase$_i$($i \geq 1$), the neuron parameters learned by the residual network during the base phase$_0$ are frozen, and a set of scaling weights ξ is applied to preserve the internal structure of the neurons [42], allowing the knowledge retention block to remember the base phase$_0$ knowledge while gradually adapting to the new data.

Specifically, assume the network parameters from the base phase are θ_{base}, with the ath layer containing B neurons. Thus, there are B neuron weights $\{W_{a,b}\}_{b=1,\ldots,B}$, denoted as W_a for simplicity. For W_a, B scaling weights are learned, denoted as ξ_a. Let X_{a-1} and X_a be the input and output of the ath layer, respectively. The scaling weights ξ_a are applied to W_a as follows:

$$X_a = (W_i \odot \xi_a) X_{a-1} \tag{3}$$

where $\odot$ indicates elementwise multiplication. Assuming there are A layers in total, the overall scaling weights can be represented as follows:

$$\xi = \{\xi_1, \xi_2, \ldots, \xi_A\}. \tag{4}$$

The final parameters of the knowledge retention block are denoted as $\xi = \theta_{\text{base}} \odot \xi$, which will be referred to as ξ hereafter for simplicity. Most of the parameters in the knowledge retention block are frozen, with only a small portion of the parameters being updated.

2) Knowledge Adaptation Blocks: To adapt to new data and learn new knowledge, a knowledge adaptation block is designed. The knowledge adaptation block is initialized from the residual network of the base phase. During the incremental phase, all parameters ψ of the knowledge adaptation block are updated to continuously learn new knowledge from new classes of data. Compared to the knowledge retention block, the knowledge adaptation block has significantly more learnable parameters.

3) Contrastive Learning Blocks: To learn the general knowledge of both old and new data in the shallow network, the contrastive learning block is added at the first level, in addition to the knowledge adaptation block, knowledge retention block, and knowledge fusion block. Specifically, at the first level, the features $x_\xi^{[1]}$ and $x_\psi^{[1]}$ extracted by the knowledge adaptation block and the knowledge retention block are first input into the contrastive learning block to obtain $x'^{[1]}_\xi$ and $x'^{[1]}_\psi$, respectively, as follows:

$$x'^{[1]}_\xi = F_\tau\left(x_\xi^{[1]}\right) \tag{5}$$

$$x'^{[1]}_\psi = F_\tau\left(x_\psi^{[1]}\right) \tag{6}$$

where F_τ represents the contrastive learning block network, which consists of fully connected layers. In each incremental phase$_i$($i \geq 1$), all its parameters τ are updated. Then, the negative cosine similarity between $x_\xi^{[1]}$ and $x'^{[1]}_\psi$ is calculated as follows:

$$d\left(x_\xi^{[1]}, x'^{[1]}_\psi\right) = -\frac{x_\xi^{[1]}}{\left|\left|x_\xi^{[1]}\right|\right|_2} \cdot \frac{x'^{[1]}_\psi}{\left|\left|x'^{[1]}_\psi\right|\right|_2} \tag{7}$$

where $||\cdot||_2$ is the L2-norm. Similarly, we can obtain

$$d\left(x_\psi^{[1]}, x'^{[1]}_\xi\right) = -\frac{x_\psi^{[1]}}{\left|\left|x_\psi^{[1]}\right|\right|_2} \cdot \frac{x'^{[1]}_\xi}{\left|\left|x'^{[1]}_\xi\right|\right|_2}. \tag{8}$$

We minimize the cosine similarity of $x_\xi^{[1]}$ and $x'^{[1]}_\psi$, $x_\psi^{[1]}$ and $x'^{[1]}_\xi$, respectively, and then take the mean to obtain L_d

$$L_d = \frac{1}{2} d\left(x_\xi^{[1]}, x'^{[1]}_\psi\right) + \frac{1}{2} d\left(x_\psi^{[1]}, x'^{[1]}_\xi\right). \tag{9}$$

Since the knowledge retention block primarily contains old knowledge and the knowledge adaptation block primarily contains new knowledge, minimizing the negative cosine similarity of features extracted from the knowledge adaptation block and the knowledge retention block can effectively maximize the similarity between old and new class knowledge. This alignment of new and old class features in the shallow network facilitates the extraction of general knowledge from both old and new data.

To prevent the model from being trapped in local optima or overfitting, additional fully connected layers are added to the contrastive learning block, and a stop-gradient operation is implemented. The specific details are shown in Fig. 3. By employing the stop-gradient operation, the update in (7) is revised to

$$d\left(x_\xi^{[1]}, x'^{[1]}_\psi\right) = -\text{stopgrad}\left(\frac{x_\xi^{[1]}}{\left|\left|x_\xi^{[1]}\right|\right|_2}\right) \cdot \frac{x'^{[1]}_\psi}{\left|\left|x'^{[1]}_\psi\right|\right|_2} \tag{10}$$

where $(x_\xi^{[1]}/||x_\xi^{[1]}||_2)$ is treated as a constant, which means the knowledge retention block does not receive gradient backpropagation, while the knowledge adaptation block does receive gradient backpropagation. Similarly, (10) and (9) become

$$d\left(x_\psi^{[1]}, x'^{[1]}_\xi\right) = -\text{stopgrad}\left(\frac{x_\psi^{[1]}}{\left|\left|x_\psi^{[1]}\right|\right|_2}\right) \cdot \frac{x'^{[1]}_\xi}{\left|\left|x'^{[1]}_\xi\right|\right|_2} \tag{11}$$

$$L_d = \frac{1}{2} d\left(\text{stopgrad}\left(x_\xi^{[1]}\right), x'^{[1]}_\psi\right) + \frac{1}{2} d\left(\text{stopgrad}\left(x_\psi^{[1]}\right), x'^{[1]}_\xi\right) \tag{12}$$

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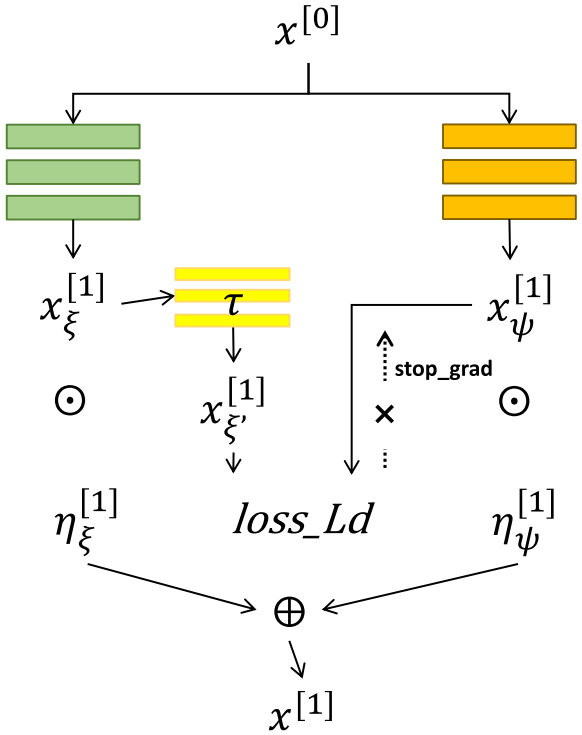


Fig. 3. Details of the contrast learning module and knowledge fusion process.

respectively. In our contrastive learning module, neither negative sample pairs are required nor do we use momentum encoders, and it does not rely on large-batch training.

4) Knowledge Fusion Blocks: Since the data ε_i and D_i from the old and new classes have different distributions, their feature spaces also differ. Simply retaining the data may overlook some important knowledge. Therefore, in addition to retaining part of the data, freezing parts of the network can preserve more effective knowledge. We design the knowledge fusion block, which uses an interpolation function at each level of the network to fuse the frozen old knowledge from the knowledge retention block with the new knowledge learned by the knowledge adaptation block. The details of the knowledge fusion process are shown in Fig. 3. Let $\eta_{\xi}^{[i]}$ and $\eta_{\psi}^{[i]}$ be the fusion weights for the knowledge retention block and the knowledge adaptation block at the ith level, respectively. The weighted knowledge fusion for the ith level is as follows:

$$x^{[i]} = x_{\xi}^{[i]} \cdot \eta_{\xi}^{[i]} + x_{\psi}^{[i]} \cdot \eta_{\psi}^{[i]}. \tag{13}$$

The weighted knowledge fusion determines the proportion of new and old knowledge combined at each level. As shown in Figs. 2 and 3, each level requires a set of fusion weights, and each incremental phase needs six such fusion weights. Since LLL involves multiple phases, a large number of fusion weights are required. Manually setting these fusion weights is cumbersome and impractical. Therefore, the fusion weights are treated as network parameters, and a global BOP is designed to automatically adjust the weights at each level for adaptive knowledge fusion.

D. Dynamic Angle Penalty Loss

As new classes emerge, the space left for new classes in the latent feature space becomes increasingly limited, leading to reduced adaptability of the model to new classes. Dynamically adjusting class margins based on the number of classes can help to reasonably allocate the feature space for both old and new data. We design an angular penalty loss for dynamic adjustment of class margins by using cosine similarity as the distance metric to measure similarity. The objective is to focus on the angle between normalized features rather than the absolute distance in the latent feature space. The data prediction process can be expressed as follows:

$$y_c = U_c^{\top} g = \|U_c\| \|g\| \cos(\alpha_c) \tag{14}$$

where $g = \Gamma(x)$, $\Gamma(\cdot)$ denotes the feature extractor, U_c represents the weights of the fully connected layer, g refers to the features extracted from the input image x, and the angle between g and w is denoted as α_c. The variable y_c is the cosine similarity between the features g and the weights U_c, which represents the probability that the image x belongs to class c. Both g and U_c are l_2-normalized to unit length.

Typically, cosine similarity is used in conjunction with cross-entropy loss to enhance the separability of features from different classes by maximizing the probability of the true class. The loss function is formulated as follows:

$$\begin{aligned} L_{\mathrm{AP}} &= -\frac{1}{N}\sum_{k=1}^{N}\log(p_n) = -\frac{1}{N}\sum_{n=1}^{N}\log\left(\frac{\exp(y_n)}{\sum_{c=1}^{C}\exp(y_c)}\right) \\ &= -\frac{1}{N}\sum_{n=1}^{N}\log\left(\frac{\exp(\|U_c\|\|g\|\cos(\alpha_n))}{\sum_{c=1}^{C}\exp(\|U_c\|\|g\|\cos(\alpha_c))}\right) \\ &= -\frac{1}{N}\sum_{n=1}^{N}\log\left(\frac{\exp(\cos(\alpha_n))}{\sum_{c=1}^{C}\exp(\cos(\alpha_c))}\right) \end{aligned} \tag{15}$$

where N is the number of training images, C is the number of classes, p_n represents the softmax probability of image x_n, and y_n represents the cosine similarity between image x_n and its ground-truth class.

In the incremental phase, as the number of classes increases, to prevent significant overlap in the feature space of new and old classes causing classification errors, a class margin Δ is introduced. This makes intraclass features more compact and interclass features more separated. On this basis, the normalized features are rescaled by a scaling factor β, which increases the relative distances between features as the number of classes increases, thus improving classification accuracy.

Since in LLL, the number of data classes continually increases, using the same class margin each time is not reasonable. Therefore, in the incremental phase, the value of Δ should increase with the number of classes. Specifically, when the number of classes is large, it is required that $\alpha_j \leq (\pi/2)$ and $\cos(\alpha_j) \geq 0$. At this point, e^x is a monotonically increasing function. To maintain the monotonicity of the function, we set $\Delta \leq 1$. Therefore, Δ is designed as follows:

$$L_{\mathrm{DAP}} = -\frac{1}{N}\sum_{n=1}^{N}\log\left(\frac{e^{(\beta(\cos(\alpha_n)-\Delta))}}{e^{(\beta(\cos(\alpha_n)-\Delta))} + \sum_{c\neq n} e^{(\beta\cos(\alpha_c))}}\right) \tag{16}$$

$$\Delta = \tanh\left(\sum_{i=0}^{P}\frac{1}{\text{class_base} + p \times \text{class_increment}}\right). \tag{17}$$

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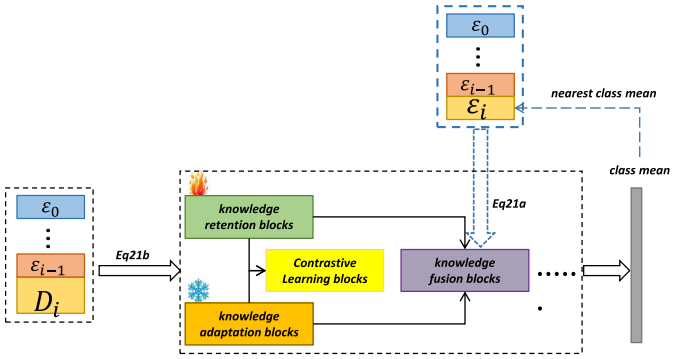


Fig. 4. Details of the BOP based on different data strategies.

In phase p, class_base represents the number of classes in the base phase and class_incremental represents the number of classes added in each incremental phase.

E. BOP Based on Different Data Strategies

To design a BOP to achieve adaptive knowledge fusion, we incorporate the fusion parameters as part of the network and adaptively optimize them based on the constraints. Specifically, all network parameters are divided into two groups for optimization: one group $[\xi_i, \psi_i, \tau_i]$ includes the parameters of the knowledge retention block, the knowledge adaptation block, and the contrastive learning module; the other group η_i consists of the parameters of the knowledge fusion block. Initially, the value of η_i is set to 0.5. Then, with η_i as a constraint, the model parameters $[\xi_i, \psi_i, \tau_i]$ are optimized to obtain an optimal set of parameters $[\xi_i^*, \psi_i^*, \tau_i^*]$. Conversely, with $[\xi_i^*, \psi_i^*, \tau_i^*]$ being fixed, η_i is updated. This process can be expressed as follows:

$$\begin{aligned} &\min_{\eta_i} L\left(\eta_i, \xi_i^*, \psi_i^*, \tau_i^*; D_{0,1,\ldots,i-1} \cup D_i\right) \\ &\text{s.t.}\left[\xi_i^*, \psi_i^*, \tau_i^*\right] = \underset{[\xi_i, \psi_i, \tau_i]}{\arg\min} \left(\eta_i, \xi_i, \psi_i, \tau_i; D_{0,1,\ldots,i-1} \cup D_i\right) \end{aligned} \tag{18}$$

where $L(\cdot)$ is the final loss function.

Due to memory limitations, it is impossible to completely retain all the data from the old classes. Therefore, the NCM is used to select a portion of the data as the retained old knowledge for the current phase, which is then replayed in the new phase. Thus, the input data for phase$_i$ is $\varepsilon_{0,1,\ldots,i-1} \cup D_i$. At this point, only a small amount of old data is retained for each class from phase$_{0,1,\ldots,i-1}$, while each class in phase$_i$ has a large amount of new data. This imbalance between the new and old class data can lead to catastrophic forgetting, causing the prediction results to be biased toward the new classes with large amounts of data.

To address this issue, a BOP based on different data strategies is designed, as detailed in Fig. 4. On the one hand, the network parameters of the knowledge adaptation block, knowledge retention block, and contrastive learning module are trained using $\varepsilon_{0,1,\ldots,i-1} \cup D_i$, with the objective of training the main parameters of the feature extraction network using all available samples, enabling the network to learn as much knowledge as possible. On the other hand, the fusion weights are trained using $\varepsilon_{0,1,\ldots,i-1} \cup \varepsilon_i$, with the objective of adjusting the fusion proportion of new and old knowledge using an equal amount of new and old class data, thereby mitigating catastrophic forgetting caused by data imbalance. This process is represented by the following equation:

$$\begin{aligned} &\min_{\eta_i} L\left(\eta_i, \xi_i^*, \psi_i^*, \tau_i^*; \varepsilon_{0,1,\ldots,i-1} \cup \varepsilon_i\right) \\ &\text{s.t.}\left[\xi_i^*, \psi_i^*, \tau_i^*.\right] = \underset{[\xi_i, \psi_i, \tau_i]}{\arg\min} \left(\eta_i, \xi_i, \psi_i, \tau_i; \varepsilon_{0,1,\ldots,i-1} \cup D_i\right). \end{aligned} \tag{19}$$

The total loss for the base phase is given as follows:

$$L_{\text{base}} = L_m + L_c \tag{20}$$

where L_{base} represents the loss of base phase, L_m denotes the loss associated with data augmentation, and L_c is the cross-entropy loss.

The total loss for the incremental phases is given as follows:

$$L_{\text{incremental}} = L_{\text{DAP}} + L_d \tag{21}$$

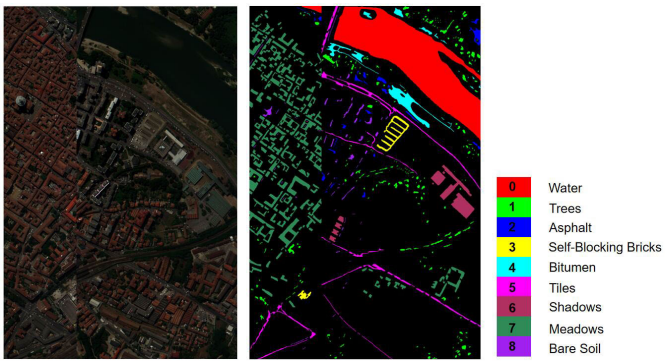
where L_{DAP} is the dynamic angle penalty loss, while L_d represents the contrastive learning loss.

In Algorithm 1, the training steps of the proposed method are summarized in the incremental phasr$_i$($i \geq 1$). Lines 1–3 involve loading data and initializing parameters, lines 4–11 involve the alternating optimization of network and knowledge fusion parameters, and lines 12–13 are used to update the retained representative instances.

Algorithm 1 Training Process (in the ith ($i \geq 1$) Phase) of LLL-AFCA

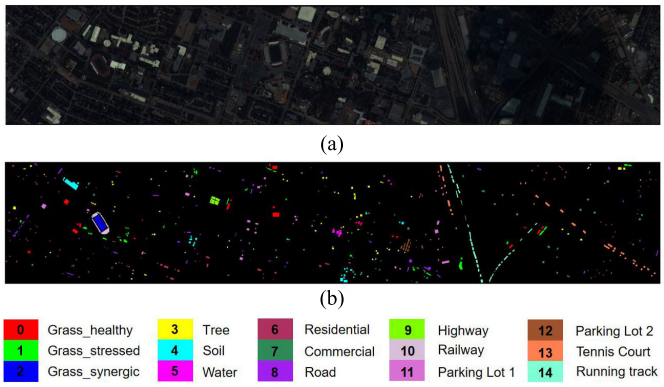
Input:
New class data D_i
Old class exemplars $\varepsilon_{0,1,\ldots i-1}$
Old parameters $[\eta_{i-1}, \xi_{i-1}, \psi_{i-1}, \tau_{i-1}]$
Base model θ_{base} or old model θ_{i-1}
Output:
The total loss $L_{incremental}$ of the model
New parameters $[\eta_i, \xi_i, \psi_i, \tau_i]$
New class exemplars ε_i
New model θ_i
1: Get D_i and load $\varepsilon_{0,1,\ldots i-1}$ from memory
Initialize $[\xi_i, \psi_i, \tau_i]$ with $[\xi_{i-1}, \psi_{i-1}, \tau_{i-1}]$
Initialize η_i with η_{i-1}
2: **for** episode in {1, . . . , E} **do**
3: **for** mini-batches in $\varepsilon_{0,1,\ldots i-1} \cup D_i$ do **do**
4: Train $[\xi_i, \psi_i, \tau_i]$ on $\varepsilon_{0,1,\ldots i-1} \cup D_i$ by Eq 19
5: **end for**
6: **for** mini-batches in $\varepsilon_{0,1,\ldots i-1} \cup \varepsilon_i$ do **do**
7: Train η_i on $\varepsilon_{0,1,\ldots i-1} \cup \varepsilon_i$ by Eq 20 (i=0) or Eq 21 (i>0)
8: **end for**
9: **end for**
10: Update exemplars ε_i by NCM
Replace $\varepsilon_{0,1,\ldots i-1}$ with $\varepsilon_{0,1,\ldots i-1} \cup \varepsilon_i$ in the memory.

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(a) (b)

Fig. 5. Pseudo-color image and ground-truth map of PC. (a) Pseudo-color image. (b) Ground-truth map.



(a)

(b)

Fig. 6. Pseudo-color image and ground-truth map of HT. (a) Pseudo-color image. (b) Ground-truth map.

IV. Experiments

A. Datasets

The proposed LLL-AFCA method was validated through experiments on four public datasets: Pavia Centre (PC), Houston2013 (HU13), Salinas (SA), and Houston2018 (HU18).

1) PC: The PC dataset was collected using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over Pavia in northern Italy. After removing noisy bands, there are 102 spectral channels covering the range from 430 to 860 nm, with a spatial resolution of 1.3 m. The spatial size is 1096 × 715 pixels, containing nine classes with a total of 7456 samples, including asphalt and shadows. The pseudo-color image of the corresponding ground truth is shown in Fig 5. Table I shows the number of samples per class in the PC dataset.
2) *HU13:* The HU dataset was acquired using the Compact Airborne Spectrographic Imager (CASI) sensor over the University of Houston campus and its surrounding areas. It has 144 spectral channels covering the range from 380 to 1050 nm, with a geometric resolution of 2.5 m, and a spatial size of 349 × 1905 pixels. It contains 15 classes with a total of 15 029 labeled samples. The pseudo-color image and its corresponding ground-truth image are shown in Fig. 6. Table II shows the number of samples per class in the Houston dataset.
3) *SA:* The SA dataset was captured using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the Salinas Valley in California, USA. The original image contains 224 spectral channels. After removing noisy bands, 204 spectral channels remain, covering the range from 400 to 2500 nm. This HIS has a spatial resolution of 3.7 m and a spectral resolution of 10 nm, with a spatial size of 512 × 217 pixels, containing 16 classes with a total of 54 129 samples. The pseudo-color image and the corresponding ground-truth image are shown in Fig. 7. Table III shows the number of samples per class in the Salinas dataset.
4) *HU18:* The Houston18 dataset was captured using the ITRES-CASI 1500 hyperspectral imaging sensor over the University of Houston campus and its neighboring urban areas. The original dataset contains 48 spectral bands after preprocessing, covering the spectral range from 380 to 1050 nm. This HIS has a spatial resolution of 1 m and a spatial size of 601 × 2384 pixels, containing 20 classes with a total of 101 500 labeled samples. The pseudo-color image and the corresponding ground-truth image are shown in Fig. 8. Table IV provides the number of samples per class in



(a) (b)

Fig. 7. Pseudo-color image and ground-truth map of SA. (a) Pseudo-color image. (b) Ground-truth map.

TABLE I
NUMBER OF SAMPLES FOR THE PC DATASET

Class	Name	Number of Samples
1	Water	824
2	Trees	820
3	Asphalt	816
4	Self-Blocking Bricks	808
5	Bitumen	808
6	Tiles	1260
7	Shadows	476
8	Meadows	824
9	Bare Soil	820
	Total	7456

TABLE II
NUMBER OF SAMPLES FOR THE HU13 DATASET

Class	Name	Number of Samples
1	Healthy grass	1053
2	Stressed grass	1064
3	Synthetic grass 2	505
4	Trees	1056
5	Soil	1056
6	Water	143
7	Residential	1072
8	Commercial	1053
9	Road	1059
10	Highway	1036
11	Railway	1054
12	Parking Lot 1	1041
13	Parking Lot 2	285
14	Tennis Court	247
15	Running Track	473
	Total	12197

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TABLE III
NUMBER OF SAMPLES FOR THE SA DATASET

Class	Name	Number of Samples
1	Brocoli green weeds 1	2009
2	Brocoli green weeds 2	3726
3	Fallow	1976
4	Fallow rough plow	1394
5	Fallow smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes untrained	11271
9	Soil vinyard develop	6203
10	Corn senesced green weeds	3278
11	Lettuce romaine 4wk	1068
12	Lettuce romaine 5wk	1927
13	Lettuce romaine 6wk	916
14	Lettuce romaine 7wk	1070
15	Vinyard untrained	7268
16	Vinyard vertical trellis	1807
	Total	54129

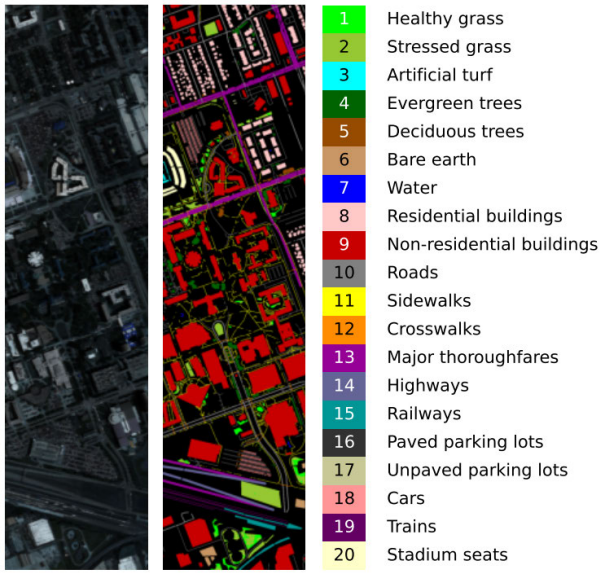
TABLE IV
NUMBER OF SAMPLES FOR THE HU18 DATASET

Class	Name	Number of Samples
1	Healthy grass	9799
2	Stressed grass	32502
3	Artificial turf	684
4	Evergreen trees	13595
5	Deciduous trees	5021
6	Bare earth	4516
7	Water	266
8	Residential buildings	39772
9	Non-residential buildings	223752
10	Roads	45866
11	Sidewalks	34029
12	Crosswalks	1518
13	Major thoroughfares	46348
14	Highways	9865
15	Railways	6937
16	Paved parking lots	11500
17	Unpaved parking lots	146
18	Cars	6547
19	Trains	5396
20	Stadium seats	6824
	Total	504856

the Houston18 dataset. On the Houston18 dataset, the performance advantage brought by data augmentation in the last method (LLL-AFCA) is partially offset due to the dataset's inherent characteristics, data distribution, and potential interfering factors, resulting in performance that may not necessarily surpass other methods. However, this augmentation strategy demonstrates better effectiveness on other datasets, typically achieving superior results in other experiments. Overall, data augmentation is not the core innovation of this algorithm, and its effectiveness varies depending on different data characteristics and task requirements.

B. Implementation Details

1) *Experimental Environment:* The proposed method was implemented on an Ubuntu 18.04 Linux operating system, equipped with an Intel Xeon processor (3.7 GHz), 64 GB of memory, and an NVIDIA TITAN RTX 24 GB GPU. The corresponding CUDA version was 10.2, and the open-source software framework used was PyTorch.
2) *Data Preprocessing:* The HSI datasets were divided into two subsets: a training set and a testing set. For each class, 30 samples were selected as the training set, and the remaining samples were used as the testing set. In the base phase, half or more of the classes from each dataset were used for training. In the incremental phase, two classes were added for training in each phase, with 10 samples retained from each old class. Specifically, PC dataset contains 9 classes. Five classes were used as the base phase data, and two classes were added in each incremental phase, resulting in two incremental phases. HU sataset contains 15 classes. Nine classes were used as the base phase data, and two classes were added in each incremental phase, resulting in three incremental phases. SA dataset contains 16 classes. Eight classes were used as the base phase data, and two classes were added in each incremental phase, resulting in four incremental phases. HU18 dataset contains 20 classes. Ten classes were used as the base phase data, and two classes were added in each incremental phase, resulting in five incremental phases. A 9×9 spatial neighborhood was selected as the input cube. Principal component analysis (PCA) was used to extract the first 15 principal components, which was to remove spectral redundancy.
3) *Architectures:* An 8-layer ResNet was adopted as the baseline network architecture. This ResNet consisted of an initial convolutional layer and three residual blocks, each containing two convolutional layers with 3×3 kernels. The number of filters started at 16 and doubled with each subsequent residual block. After the three residual blocks, an average pooling layer was added for feature compression, followed by a fully connected layer for output.
4) *Hyperparameters and Configuration:* We redesigned the experimental part by dividing the dataset into three parts, with 30 samples of each class as the training set, 10 samples of each class as the validation set, and the remaining samples as the test set to determine the model hyperparameters through the validation set.
5) *Evaluation Metrics:* Our method was evaluated using three metrics: overall accuracy (OA), which is calculated



(a) (b)

Fig. 8. Pseudo-color image and ground-truth map of HU. (a) Pseudo-color image. (b) Ground-truth map.

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TABLE V
CLASSIFICATION RESULTS OF LLL-AFCA METHOD ON THE PC DATASET (%)

Class	Base phase0 (5 classes)	Incremental phase1 (7 classes)	Incremental phase2 (9 classes)
1	100.000 ± 0.000	99.308 ± 1.981	99.746 ± 0.439
2	95.177 ± 3.320	94.574 ± 1.486	89.299 ± 5.935
3	96.585 ± 1.035	91.759 ± 7.927	96.621 ± 3.105
4	98.158 ± 1.703	94.524 ± 6.688	96.889 ± 3.569
5	93.049 ± 3.751	92.450 ± 3.634	86.654 ± 8.077
6		99.031 ± 1.625	87.175 ± 7.233
7		93.204 ± 5.444	87.057 ± 5.449
8			98.767 ± 0.697
9			97.561 ± 1.776
OA	**98.917 ± 0.359**	**97.796 ± 1.457**	**96.958 ± 0.768**
AA	96.574 ± 0.799	94.936 ± 1.660	93.252 ± 1.400
Kappa	97.095 ± 0.707	95.928 ± 2.343	95.413 ± 0.732

TABLE VI
CLASSIFICATION RESULTS OF LLL-AFCA METHOD ON THE HU13 DATASET (%)

Class	Base phase0 (9 classes)	Incremental phase1 (11 classes)	Incremental phase2 (13 classes)	Incremental phase3 (15 classes)
1	96.421 ± 4.398	96.413 ± 2.445	97.224 ± 1.726	97.199 ± 1.978
2	98.644 ± 0.723	97.557 ± 3.683	96.993 ± 3.987	96.642 ± 4.172
3	99.955 ± 0.101	99.940 ± 0.105	99.700 ± 0.367	99.535 ± 0.438
4	96.985 ± 2.926	96.886 ± 2.101	95.692 ± 3.074	95.857 ± 2.633
5	99.959 ± 0.105	100.000 ± 0.000	99.984 ± 0.035	99.975 ± 0.056
6	96.780 ± 5.417	97.729 ± 4.341	98.000 ± 4.217	97.559 ± 4.185
7	93.667 ± 2.711	85.679 ± 6.497	84.806 ± 5.519	86.107 ± 6.409
8	90.675 ± 3.540	82.331 ± 5.151	77.521 ± 3.942	78.443 ± 5.457
9	96.637 ± 2.053	71.669 ± 7.749	65.581 ± 6.886	67.062 ± 5.604
10		98.079 ± 2.609	91.529 ± 6.124	90.994 ± 5.951
11		96.232 ± 3.645	88.996 ± 6.015	86.788 ± 6.642
12			95.436 ± 3.499	91.039 ± 5.458
13			92.369 ± 2.629	86.241 ± 4.562
14				100.000 ± 0.000
15				100.000 ± 0.000
OA	**96.425 ± 0.714**	**92.233 ± 1.519**	**90.148 ± 1.109**	**90.345 ± 1.006**
AA	96.636 ± 0.614	92.956 ± 1.520	91.064 ± 0.991	91.563 ± 0.906
Kappa	95.932 ± 0.812	91.394 ± 1.683	89.250 ± 1.209	89.559 ± 1.088

by dividing the number of correctly classified samples by the total number of testing samples; average accuracy (AA), which is the AA of all classes; and the Kappa coefficient, which is used to evaluate the agreement between the model's predictions and the actual classification results.

C. Experimental Results

To ensure the reliability of the experimental results, ten experiments with randomly selected training samples were conducted, and the mean and standard deviation were reported. The classification results of our method on the four datasets are shown in Tables V–VIII.

On the PC dataset, our method achieved OAs of 98.917%, 97.796%, and 96.958% at each phase, respectively. Since it is not possible to retain all data in the new phase, the model partially forgets old knowledge, resulting in a decrease in OA of 1.121% and 0.838% in the current phase compared to the previous phase. On the HU13 dataset, the proposed method achieved OAs of 96.425%, 92.233%, 90.148%, and 90.345% at each phase, respectively. The OA in the current phase decreased by 4.129%, 2.085%, and -0.197% compared to the previous phase. On the SA dataset, the proposed method achieved OAs of 99.562%, 98.736%, 98.482%, 98.262%, and 92.354% at each phase, respectively. The OA in the current phase decreased by 0.826%, 0.254%, 0.220%, and 5.908% compared to the previous phase. On the HU18 dataset, the proposed method achieved OAs of 78.600%, 63.716%, 59.754%, 58.168%, 53.376%, and 53.114% at each phase, respectively. The OA in the current phase decreased by 14.884%, 3.962%, 1.586%, 4.792%, and 0.262% compared to the previous phase.

Catastrophic forgetting of old knowledge is a major challenge in LLL. The experimental results show that on the PC dataset, the classification performance of LLL-AFCA remains relatively stable during the incremental phases. On the HU13 dataset, the level of forgetting decreases over the incremental phases, and in the third incremental phase, the OA increased by 0.197% compared to the previous phase. This indicates that the model not only retains old knowledge while learning new knowledge, but old knowledge also aids in learning new knowledge. This demonstrates that the model achieves both retention of old knowledge and adaptation to new knowledge. On the SA dataset, the model is less affected by catastrophic forgetting, indicating that the overall classification accuracy remains relatively stable and is less prone to catastrophic forgetting.

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TABLE VII
CLASSIFICATION RESULTS OF LLL-AFCA METHOD ON THE SS DATASET (%)

Class	Base phase0 (8 classes)	Incremental phase1 (10 classes)	Incremental phase2 (12 classes)	Incremental phase3 (14 classes)	Incremental phase4 (16 classes)
1	99.990 ± 0.032	99.788 ± 0.654	99.788 ± 0.671	99.965 ± 0.112	99.955 ± 0.084
2	100.000 ± 0.000	99.778 ± 0.287	99.924 ± 0.240	99.922 ± 0.178	99.862 ± 0.244
3	99.959 ± 0.130	99.882 ± 0.225	99.918 ± 0.165	99.902 ± 0.175	99.265 ± 1.579
4	99.765 ± 0.260	99.428 ± 0.309	99.003 ± 0.647	99.120 ± 0.524	99.362 ± 0.301
5	96.813 ± 1.262	97.477 ± 1.090	97.232 ± 2.130	96.628 ± 2.984	95.532 ± 4.032
6	99.924 ± 0.173	99.516 ± 0.883	99.486 ± 0.812	99.519 ± 0.799	99.460 ± 0.695
7	99.992 ± 0.014	99.924 ± 0.110	99.791 ± 0.235	99.814 ± 0.430	99.634 ± 0.758
8	99.634 ± 0.582	97.747 ± 1.136	97.161 ± 1.402	96.491 ± 1.061	73.765 ± 6.911
9		99.971 ± 0.072	100.000 ± 0.000	99.997 ± 0.010	99.990 ± 0.020
10		95.788 ± 2.762	93.993 ± 2.836	93.827 ± 2.755	94.757 ± 2.139
11			99.027 ± 0.869	98.295 ± 1.940	97.765 ± 3.500
12			99.942 ± 0.183	98.672 ± 1.440	98.651 ± 2.868
13				99.639 ± 0.739	99.312 ± 1.108
14				99.808 ± 0.256	99.769 ± 0.188
15					90.019 ± 3.168
16					98.829 ± 0.683
OA	**99.562 ± 0.262**	**98.736 ± 0.467**	**98.482 ± 0.310**	**98.262 ± 0.368**	**92.354 ± 1.056**
AA	99.510 ± 0.189	98.930 ± 0.353	98.772 ± 0.255	98.685 ± 0.299	96.620 ± 0.407
Kappa	99.453 ± 0.327	98.519 ± 0.546	98.257 ± 0.355	98.029 ± 0.417	91.515 ± 1.157

TABLE VIII
CLASSIFICATION RESULTS OF LLL-AFCA METHOD ON THE HU18 DATASET (%)

Class	Base phase0 (10 classes)	Incremental phase1 (12 classes)	Incremental phase2 (14 classes)	Incremental phase3 (16 classes)	Incremental phase4 (18 classes)	Incremental phase5 (20 classes)
1	85.942 ± 5.298	82.736 ± 6.586	82.877 ± 6.419	83.392 ± 5.291	84.431 ± 4.995	84.757 ± 4.441
2	76.205 ± 5.460	70.281 ± 5.029	69.325 ± 6.014	71.570 ± 5.227	72.336 ± 5.375	72.219 ± 4.995
3	99.694 ± 0.408	99.740 ± 0.559	99.664 ± 0.602	99.755 ± 0.463	99.801 ± 0.323	99.862 ± 0.244
4	95.956 ± 1.766	91.953 ± 2.503	92.339 ± 3.402	93.852 ± 2.362	94.301 ± 2.275	94.566 ± 2.441
5	89.379 ± 1.646	86.536 ± 3.897	86.548 ± 2.793	86.309 ± 2.610	86.277 ± 2.439	87.331 ± 2.893
6	97.176 ± 2.414	95.615 ± 2.975	95.664 ± 3.159	96.545 ± 2.414	96.806 ± 2.309	96.752 ± 2.553
7	99.958 ± 0.134	99.407 ± 1.876	99.873 ± 0.402	99.873 ± 0.402	99.788 ± 0.670	99.831 ± 0.409
8	81.229 ± 3.579	67.228 ± 8.846	70.285 ± 6.062	64.794 ± 11.642	68.884 ± 9.240	66.822 ± 8.267
9	76.417 ± 3.618	65.555 ± 5.943	62.882 ± 5.860	58.899 ± 6.874	47.906 ± 9.338	45.987 ± 9.402
10	78.565 ± 2.795	82.736 ± 6.586	25.294 ± 4.238	21.990 ± 4.202	22.550 ± 3.892	22.680 ± 4.627
11		70.281 ± 5.029	33.254 ± 5.036	37.650 ± 5.865	34.909 ± 6.209	34.452 ± 6.165
12		99.740 ± 0.559	41.284 ± 6.307	36.438 ± 7.591	35.470 ± 8.765	34.469 ± 6.718
13			55.082 ± 4.239	46.268 ± 5.945	43.494 ± 6.921	41.674 ± 5.459
14			89.948 ± 4.788	87.603 ± 6.722	87.385 ± 6.323	83.790 ± 6.279
15				99.186 ± 0.544	98.578 ± 1.488	96.107 ± 2.461
16				95.179 ± 2.222	83.753 ± 6.935	85.296 ± 6.407
17					100.000 ± 0.000	99.828 ± 0.363
18					92.897 ± 2.817	88.128 ± 3.321
19						96.541 ± 2.545
20						97.848 ± 1.733
OA	**78.600 ± 2.103**	**63.716 ± 2.847**	**59.754 ± 2.762**	**58.168 ± 2.503**	**53.376 ± 3.580**	**53.114 ± 3.584**
AA	88.052 ± 0.955	76.421 ± 1.326	71.737 ± 1.083	73.706 ± 0.984	74.976 ± 0.968	76.447 ± 0.886
Kappa	68.369 ± 2.648	52.344 ± 2.786	50.341 ± 2.678	49.900 ± 2.279	46.035 ± 3.006	46.379 ± 2.983

Extensive experiments have shown that the proposed method ensures the learning of new data without forgetting the previously acquired knowledge. Additionally, the proposed method maintains excellent classification results in the final classification task.

D. Comparison With the State-of-the-Art Methods

To validate the effectiveness of LLL-AFCA, we compared it with the following LLL methods, including LUCIR [18], iCaRL [19], AANets [20], CILKD [14], DCPRN [17], and three methods related to LLL-AFCA. In order to ensure the fairness of comparison, we selected the same experimental settings as ours in all comparative experiments. The OA results of the comparative experiments at each phase are summarized in Tables IX–XII.

On the PC, HU13, SA, and HU18 datasets, the LLL-AFCA method achieved higher OA at each stage compared to other methods, significantly improving classification results.

Specifically, on the PC dataset, in the base phase, compared to the other methods, the OA increased by 0.009%, 0.175%, and 0.041%. In the first incremental phase, compared to other methods, the OA increased by 0.91%, 0.126%, 0.112%, 0.922%, and 7.902%. In the second incremental phase, compared to other methods, the OA increased by 9.924%, 3.571%, 2.53%, 0.104%, and 1.29%.

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TABLE IX
COMPARATIVE EXPERIMENTAL RESULTS ON THE PC DATASET (%)

Method	Base phase0	Incremental phase1	Incremental phase2
LUCIR	98.978 ± 0.369	96.886 ± 0.955	87.034 ± 4.475
iCaRL	98.978 ± 0.369	97.670 ± 0.653	93.387 ± 2.041
AA nets	98.978 ± 0.369	97.684 ± 0.467	94.428 ± 0.897
CILKD	98.812 ± 0.352	96.874 ± 0.535	96.854 ± 0.650
DCPRN	98.946 ± 0.166	89.894 ± 7.451	95.668 ± 2.208
LLL-A	98.621 ± 0.244	97.541 ± 0.641	93.619 ± 2.726
LLL-CA	98.621 ± 0.244	97.634 ± 0.649	96.868 ± 0.672
LLL-AFCA	**98.987 ± 0.359**	**97.796 ± 1.457**	**96.958 ± 0.768**

TABLE X
COMPARATIVE EXPERIMENTAL RESULTS ON THE HU13 DATASET (%)

Method	iteration_0	iteration_1	iteration_2	iteration_3
LUCIR	94.491 ± 1.329	80.339 ± 2.897	78.965 ± 3.490	83.657 ± 3.030
iCaRL	94.491 ± 1.329	87.803 ± 2.733	87.372 ± 2.477	87.995 ± 2.398
AA nets	94.491 ± 1.329	88.948 ± 2.228	87.937 ± 2.207	88.492 ± 1.748
CILKD	91.352 ± 1.750	87.549 ± 0.385	81.711 ± 1.298	83.047 ± 1.017
DCPRN	95.347 ± 1.399	87.242 ± 2.941	85.311 ± 1.855	83.971 ± 1.713
LLL-A	93.131 ± 1.240	87.492 ± 2.087	85.738 ± 2.023	87.375 ± 2.081
LLL-CA	93.131 ± 1.240	87.860 ± 1.867	86.143 ± 2.210	87.645 ± 1.510
LLL-AFCA	**96.425 ± 0.714**	**92.233 ± 1.519**	**90.148 ± 1.109**	**90.345 ± 1.006**

TABLE XI
COMPARATIVE EXPERIMENTAL RESULTS ON THE SA DATASET (%)

Method	iteration_0	iteration_1	iteration_2	iteration_3	iteration_4
LUCIR	99.275 ± 0.488	94.235 ± 2.341	93.695 ± 1.885	94.979 ± 1.339	86.227 ± 2.087
iCaRL	99.275 ± 0.488	96.414 ± 1.540	97.264 ± 0.756	97.400 ± 0.744	89.492 ± 1.384
AA nets	99.275 ± 0.488	98.204 ± 0.724	97.932 ± 0.436	97.596 ± 0.633	91.377 ± 1.645
CILKD	99.385 ± 0.362	97.021 ± 0.655	95.914 ± 0.605	96.116 ± 0.956	77.282 ± 1.849
DCPRN	99.734 ± 0.199	97.129 ± 0.799	97.087 ± 0.902	96.098 ± 0.819	82.897 ± 3.124
LLL-A	99.407 ± 0.339	98.221 ± 0.508	98.132 ± 0.453	98.091 ± 0.535	91.363 ± 0.751
LLL-CA	99.407 ± 0.339	98.374 ± 0.397	98.235 ± 0.466	98.167 ± 0.555	91.535 ± 0.748
LLL-AFCA	**99.562 ± 0.262**	**98.736 ± 0.467**	**98.482 ± 0.310**	**98.262 ± 0.368**	**92.354 ± 1.056**

TABLE XII
COMPARATIVE EXPERIMENTAL RESULTS ON THE HU18 DATASET (%)

Method	iteration_0	iteration_1	iteration_2	iteration_3	iteration_4	iteration_5
LUCIR	75.692 ± 3.293	62.516 ± 3.981	56.473 ± 3.915	56.513 ± 3.441	53.578 ± 2.504	51.013 ± 4.094
iCaRL	75.692 ± 3.293	60.425 ± 3.379	56.039 ± 3.643	57.644 ± 3.059	53.892 ± 2.830	51.413 ± 4.585
AAnets	75.692 ± 3.293	64.618 ± 2.646	59.247 ± 2.649	**59.763 ± 3.317**	54.506 ± 3.932	51.195 ± 4.747
CILKD	73.775 ± 4.175	45.011 ± 3.603	51.068 ± 3.807	47.527 ± 4.244	40.567 ± 1.457	40.975 ± 2.015
DCPRN	78.017 ± 2.252	61.316 ± 3.629	59.653 ± 3.557	50.654 ± 2.941	46.991 ± 1.757	36.488 ± 3.158
LLL-A	75.692 ± 3.293	63.087 ± 2.504	59.656 ± 2.797	59.144 ± 3.272	54.000 ± 2.758	51.359 ± 3.449
LLL-CA	75.692 ± 3.293	**66.234 ± 3.500**	59.671 ± 3.232	59.824 ± 2.971	**54.590 ± 2.813**	51.473 ± 3.269
LLL-AFCA	**78.600 ± 2.103**	63.716 ± 2.847	**59.754 ± 2.762**	58.168 ± 2.503	53.376 ± 3.580	**53.114 ± 3.584**

On the HU13 dataset, in the base phase, compared to the first four methods, the OA increased by 1.934%, 5.073%, and 1.078%; in the first incremental phase, compared to other methods, the OA increased by 11.894%, 4.430%, 3.285%, 4.684%, and 4.991%; in the second incremental phase, compared to other methods, the OA increased by 11.183%, 2.776%, 2.211%, 8.437%, and 4.837%; and in the third phase, compared to other methods, the OA increased by 6.688%, 2.350%, 1.853%, 7.298%, and 6.374%.

On the SA dataset, in the base phase, compared to the first four methods, the OA increased by 0.287% and 0.177%; in the first incremental phase, compared to other methods, the OA increased by 4.501%, 2.322%, 0.532%, 1.715%, and 1.607%; in the second incremental phase, compared to other methods, the OA increased by 4.787%, 1.218%, 0.550%, 2.568%, and 1.395%; in the third incremental phase, compared to other methods, the OA increased by 3.283%, 0.862%, 0.666%, 2.146%, and 2.164%; and in the fourth incremental phase, compared to other methods, the OA increased by 6.127%, 2.862%, 0.977%, 15.072%, and 9.457%.

On the HU18 dataset, in the base phase, compared to the first four methods, the OA increased by 2.908%, 4.825%, and 0.583%; in the first incremental phase, compared to other methods, the OA increased by 1.200%, 3.291%, 18.705%, and 2.4%; in the second incremental phase, compared to other methods, the OA increased by 3.281%, 3.715%, 0.507%, 8.686%, and 0.101%; in the third incremental phase, compared to other methods, the OA increased by 1.655%, 0.524%,

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TABLE XIII
COMPARATIVE EXPERIMENTAL RESULTS ON THE PC DATASET (%)

Class	LUCIR	iCaRL	AA nets	DCPRN	CILKD	LLL-A	LLL-CA	LLL-AFCA
1	94.835 ± 7.707	99.454 ± 1.092	99.875 ± 0.129	98.030 ± 1.590	**99.984 ± 0.020**	99.908 ± 0.179	99.744 ± 0.481	99.746 ± 0.439
2	78.451 ± 11.023	84.345 ± 9.064	88.075 ± 8.076	92.130 ± 2.510	95.856 ± 3.229	87.023 ± 8.517	86.957 ± 4.169	**89.299 ± 5.935**
3	89.654 ± 5.365	87.549 ± 13.995	93.641 ± 5.136	94.300 ± 1.370	82.564 ± 4.620	95.569 ± 3.230	96.346 ± 2.799	**96.621 ± 3.105**
4	78.550 ± 20.337	83.556 ± 14.068	90.670 ± 10.946	83.530 ± 4.410	65.132 ± 3.375	93.209 ± 8.612	93.947 ± 8.177	**96.889 ± 3.569**
5	74.365 ± 21.153	75.150 ± 15.808	89.954 ± 2.987	81.080 ± 4.880	97.922 ± 1.751	87.144 ± 4.787	89.619 ± 6.359	**86.654 ± 8.077**
6	49.780 ± 13.088	64.989 ± 10.849	81.870 ± 9.333	90.020 ± 3.160	86.319 ± 4.075	79.483 ± 5.293	79.643 ± 5.987	**87.175 ± 7.233**
7	79.629 ± 14.635	82.821 ± 19.627	88.041 ± 5.084	88.890 ± 1.450	91.214 ± 8.922	86.496 ± 4.525	88.962 ± 4.618	**87.057 ± 5.449**
8	75.972 ± 14.194	76.028 ± 21.781	90.439 ± 5.134	97.880 ± 0.370	99.712 ± 0.395	90.244 ± 8.531	96.042 ± 2.705	**98.767 ± 0.697**
9	70.399 ± 24.071	81.712 ± 11.457	92.972 ± 2.902	99.060 ± 0.290	97.431 ± 2.466	93.346 ± 5.489	97.363 ± 2.009	**97.561 ± 1.776**
OA	83.208 ± 5.821	86.988 ± 7.463	93.971 ± 1.844	95.668 ± 2.208	96.854 ± 0.650	93.619 ± 2.726	95.568 ± 0.672	**96.958 ± 0.768**
AA	76.848 ± 6.481	81.734 ± 7.614	90.615 ± 1.511	91.657 ± 2.877	90.682 ± 1.359	90.269 ± 1.987	92.069 ± 1.223	**93.252 ± 1.400**
Kappa	76.975 ± 7.638	82.070 ± 9.885	91.533 ± 2.549	93.920 ± 2.975	95.548 ± 0.916	91.063 ± 3.747	93.741 ± 0.934	**95.413 ± 0.732**

TABLE XIV
COMPARATIVE EXPERIMENTAL RESULTS ON THE HU13 DATASET (%)

Class	LUCIR	iCaRL	AA nets	DCPRN	CILKD	LLL-A	LLL-CA	LLL-AFCA
1	93.653 ± 4.611	**94.513 ± 5.178**	94.161 ± 5.659	93.550 ± 1.820	88.087 ± 8.942	93.038 ± 5.025	93.571 ± 4.055	92.383 ± 6.344
2	93.382 ± 6.011	92.214 ± 5.924	93.415 ± 7.867	**95.820 ± 1.630**	88.626 ± 9.796	92.386 ± 6.604	92.672 ± 6.128	94.877 ± 5.222
3	97.781 ± 2.057	98.126 ± 1.282	99.235 ± 0.559	99.430 ± 0.180	97.343 ± 3.035	99.505 ± 0.387	**99.550 ± 0.374**	98.171 ± 1.694
4	92.265 ± 2.978	91.170 ± 4.212	92.364 ± 3.457	**96.120 ± 0.980**	90.912 ± 9.554	94.028 ± 3.055	93.418 ± 1.654	94.498 ± 3.333
5	99.984 ± 0.052	**99.992 ± 0.026**	99.802 ± 0.395	97.650 ± 1.410	88.376 ± 3.422	99.761 ± 0.644	99.893 ± 0.264	99.827 ± 0.253
6	96.644 ± 4.808	94.712 ± 5.384	96.102 ± 4.526	94.070 ± 1.980	88.407 ± 13.242	96.814 ± 4.921	**96.949 ± 4.379**	95.119 ± 5.039
7	79.435 ± 8.713	77.310 ± 6.606	77.254 ± 9.469	67.100 ± 3.750	**82.789 ± 5.959**	80.024 ± 6.852	79.604 ± 6.367	78.506 ± 6.554
8	70.305 ± 4.947	69.283 ± 6.014	69.012 ± 8.050	58.440 ± 2.970	**89.315 ± 9.188**	70.189 ± 4.415	69.193 ± 6.275	75.140 ± 8.682
9	73.830 ± 5.400	66.277 ± 7.178	71.203 ± 6.333	75.690 ± 2.160	**80.448 ± 8.114**	69.468 ± 8.031	68.707 ± 7.947	68.527 ± 7.446
10	45.146 ± 17.487	84.845 ± 10.143	81.211 ± 7.265	79.310 ± 3.930	78.721 ± 10.663	81.604 ± 7.228	81.195 ± 10.914	**92.615 ± 5.021**
11	75.353 ± 12.884	88.506 ± 4.202	87.021 ± 4.929	67.590 ± 4.490	**94.714 ± 3.692**	86.954 ± 6.359	87.876 ± 8.480	88.755 ± 4.977
12	70.258 ± 11.065	**92.735 ± 5.037**	89.659 ± 7.691	81.760 ± 3.530	65.532 ± 7.121	87.190 ± 7.164	92.020 ± 5.239	88.354 ± 7.088
13	81.367 ± 8.127	85.399 ± 4.470	85.444 ± 6.261	91.870 ± 1.620	63.399 ± 13.515	86.378 ± 4.650	85.558 ± 4.882	86.401 ± 4.307
14	98.417 ± 2.571	99.799 ± 0.331	99.573 ± 0.428	98.470 ± 0.340	65.665 ± 8.612	99.874 ± 0.178	**99.925 ± 0.238**	99.648 ± 0.295
15	99.746 ± 0.749	**100.000 ± 0.000**	99.905 ± 0.201	99.750 ± 0.170	66.714 ± 13.075	99.968 ± 0.067	99.984 ± 0.050	99.968 ± 0.100
OA	82.075 ± 3.128	87.408 ± 1.678	87.343 ± 1.544	83.971 ± 1.713	80.116 ± 2.253	87.375 ± 2.081	87.645 ± 1.510	**88.826 ± 1.555**
AA	84.504 ± 2.517	88.992 ± 1.518	89.024 ± 1.380	86.440 ± 1.164	81.937 ± 2.287	89.146 ± 1.747	89.341 ± 1.359	**90.186 ± 1.385**
Kappa	80.635 ± 3.373	86.388 ± 1.813	86.320 ± 1.664	82.704 ± 1.838	78.498 ± 2.438	86.353 ± 2.247	86.643 ± 1.633	**87.922 ± 1.679**

10.641%, and 7.514%; in the fourth incremental phase, compared to other methods, the OA increased by 12.809% and 6.385%; and in the fifth incremental phase, compared to other methods, the OA increased by 2.101%, 1.701%, 1.919%, 12.139%, and 16.626%.

The experiments demonstrate that the LLL-AFCA method offers significant advantages over other methods. Especially in the final phase, it shows a substantial improvement compared to other methods, indicating that the model not only retains old knowledge and avoids catastrophic forgetting but also adapts to new knowledge, enabling effective learning of new knowledge.

The final phase comparison of classification results for each class is summarized in Tables XIII–XV, and the corresponding classification maps are shown in Figs. 9–11. Each row represents the classification map of each method at a certain stage. On all four datasets, the final average OA of the LLL-AFCA method is higher than that of other methods. Specifically, on the PC, HU, and SA datasets, the OA of 2, 5, and 7 classes, respectively, is higher than that of other methods. This further demonstrates the effectiveness of the LLL-AFCA method.

E. Ablation Experiments

LLL-AFCA uses a single residual network as the baseline architecture and includes the following key modules.

1) Only adding the class-related angular penalty loss module (LLL-A).
2) Adding the contrastive learning module and class-related angular penalty loss module simultaneously (LLL-CA).
3) Adding a data augmentation module (M) based on (2) (LLL-AFCA).

To validate the effectiveness of these modules, ablation experiments were conducted separately on the four datasets, with the relevant classification results presented in Tables XVI–XIX. In the base phase, a single residual network was used. In the incremental phase, the same residual network structure was duplicated on the basis of the single residual network, making the network a dual-branch architecture. The network performs adaptive knowledge fusion at the end of each residual block.

After adding the class-related angular penalty loss module for dynamic class margin adjustment to the dual-branch architecture (Baseline), the OA for the PC dataset increased by 7.590% and 23.376% in the first and second incremental phases, respectively. The OA for the HU13 dataset increased by 7.615%, 13.233%, and 15.212% in the first to third incremental phases, respectively. The OA for the SA dataset increased by 6.352%, 9.387%, 10.853%, and 4.125% in the first to fourth incremental phases, respectively. For the HU18 dataset, the OA decreased by 2.368% in the first incremental

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TABLE XV
COMPARATIVE EXPERIMENTAL RESULTS ON THE SA DATASET (%)

Class	LUCIR	iCaRL	AA nets	DCPRN	CILKD	LLL-A	LLL-CA	LLL-AFCA
1	99.611 ± 0.865	97.660 ± 3.160	99.591 ± 0.536	**100.000± 0.000**	99.565 ± 0.727	99.616 ± 1.144	99.909 ± 0.270	99.368 ± 1.512
2	99.878 ± 0.273	99.640 ± 0.554	99.954 ± 0.113	99.590 ± 0.150	99.932 ± 0.108	**100.000 ± 0.000**	99.976 ± 0.077	99.786 ± 0.228
3	98.150 ± 2.682	99.820 ± 0.309	99.347 ± 0.940	96.900 ± 1.080	99.748 ± 0.266	99.455 ± 1.008	**99.851 ± 0.263**	99.378 ± 0.903
4	98.116 ± 1.097	98.519 ± 0.543	98.277 ± 1.091	98.860 ± 0.470	**99.282 ± 0573**	99.025 ± 0.598	98.820 ± 0.783	99.230 ± 0.896
5	96.424 ± 2.418	96.261 ± 3.357	96.590 ± 0.869	**98.250 ± 0.470**	95.619 ± 1.730	97.134 ± 1.272	97.474 ± 1.138	97.375 ± 1.008
6	98.908 ± 1.299	98.585 ± 1.816	99.725 ± 0.375	**99.990 ± 0.010**	98.766 ± 1.392	99.600 ± 0.570	99.860 ± 0.256	99.725 ± 0.489
7	99.420 ± 0.690	99.837 ± 0.189	99.575 ± 0.331	91.300 ± 0.910	99.738 ± 0542	99.803 ± 0.251	**99.859 ± 0.330**	99.682 ± 0.376
8	67.776 ± 15.007	74.060 ± 8.643	70.756 ± 5.153	30.000 ± 4.900	74.316 ± 9.373	69.222 ± 7.055	67.662 ± 4.320	**75.654 ± 4.941**
9	99.760 ± 0.438	99.932 ± 0.129	99.879 ± 0.248	99.170 ± 0.200	99.600 ± 0.350	99.882 ± 0.243	99.956 ± 0.090	**99.998 ± 0.005**
10	90.289 ± 3.643	92.303 ± 3.650	**94.276 ± 2.609**	85.200 ± 1.490	91.773 ± 4.765	94.002 ± 3.559	94.110 ± 2.598	92.475 ± 4.515
11	96.946 ± 2.828	98.593 ± 1.115	98.430 ± 1.224	98.660 ± 0.670	98.719 ± 0.583	**99.249 ± 0.901**	99.210 ± 0.818	98.699 ± 0.760
12	97.538 ± 3.257	99.383 ± 1.165	99.199 ± 1.103	99.630 ± 0.170	99.483 ± 0.664	99.815 ± 0.236	**99.879 ± 0.184**	99.821 ± 0.434
13	97.822 ± 1.765	99.549 ± 0.516	99.582 ± 0.422	99.590 ± 0.090	99.740 ± 0.301	99.695 ± 0.357	99.707 ± 0.665	**99.921 ± 0.169**
14	96.058 ± 3.627	97.683 ± 3.833	99.404 ± 0.715	95.040 ± 1.300	99.087 ± 0.775	99.500 ± 0.578	**99.740 ± 0.311**	98.471 ± 2.828
15	68.500 ± 15.669	89.109 ± 4.201	89.802 ± 2.105	**97.020 ± 0.580**	83.765 ± 6.359	85.800 ± 5.621	90.718 ± 2.221	87.698 ± 4.586
16	80.782 ± 19.685	95.706 ± 3.486	98.829 ± 0.592	97.950 ± 0.640	97.884 ± 0.971	98.531 ± 0.671	**98.981 ± 0.826**	97.912 ± 0.702
OA	87.055 ± 2.419	91.324 ± 1.274	91.518 ± 1.132	82.897 ± 3.124	91.377 ± 1.645	91.363 ± 0.807	91.535 ± 0.748	**92.401 ± 0.780**
AA	92.874 ± 1.496	96.040 ± 0.352	96.451 ± 0.421	92.946 ± 1.140	96.064 ± 0.646	96.271 ± 0.486	**96.607 ± 0.227**	96.575 ± 0.411
Kappa	85.623 ± 2.666	91.041 ± 1.398	90.820 ± 1.244	81.194 ± 3.392	90.430 ± 1.807	89.950 ± 1.480	90.427 ± 0.825	**91.561 ± 0.861**

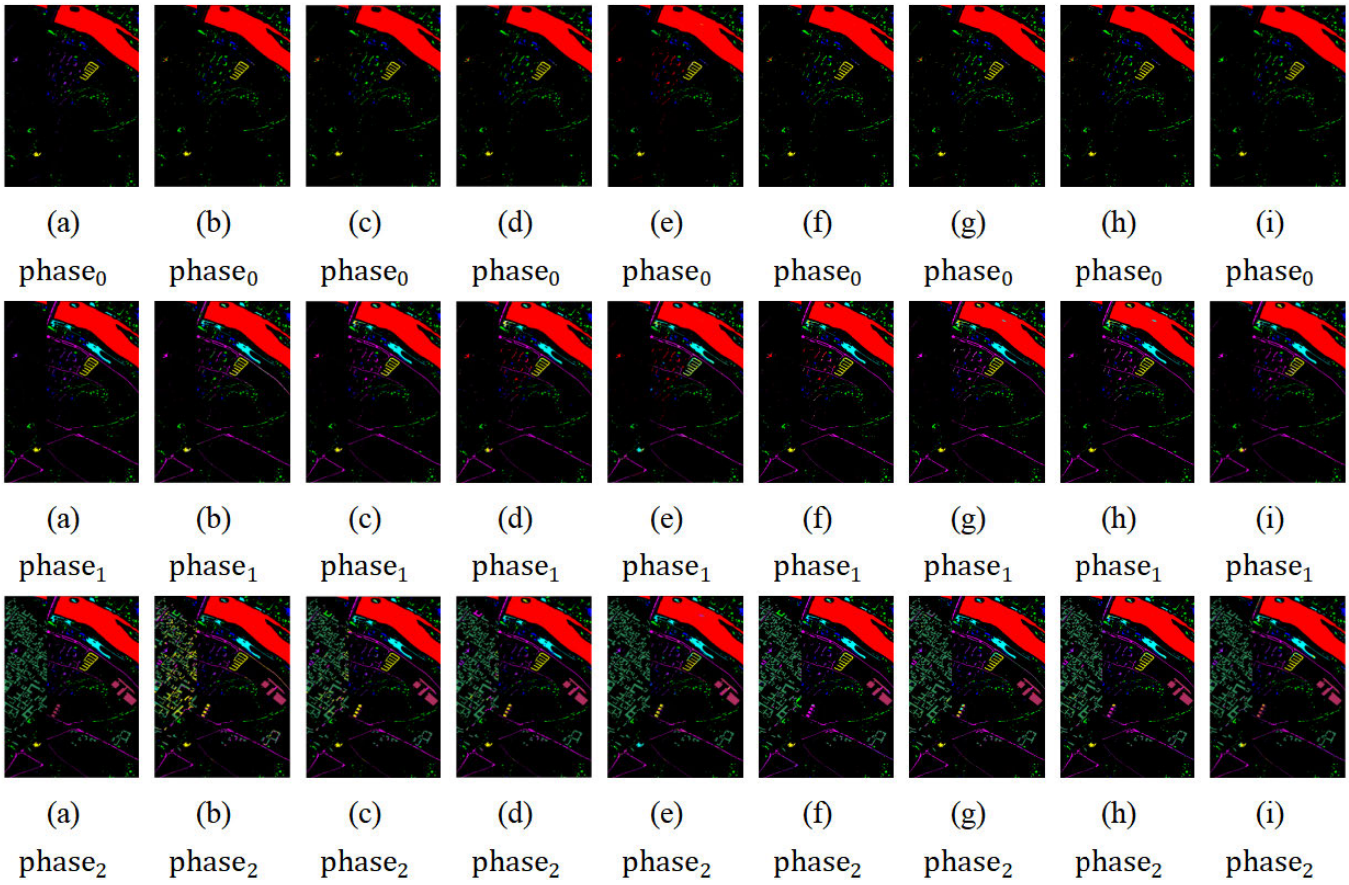


Fig. 9. Class incremental tasks classification results map of the comparative experiment on the PC dataset. (a) Ground truth. (b) LUCIR. (c) iCaRl. (d) AANet. (e) CILKD. (f) LLL-C. (g) LLL-A. (h) LLL-CA. (i) LLL-AFCA.

TABLE XVI
ABLATION EXPERIMENTS OF LLL-AFCA METHOD ON THE PC DATASET (%)

	A	C	M	Base phase0	Incremental phase1	Incremental phase2
Baseline				98.662 ± 0.588	89.951 ± 2.788	70.243 ± 8.191
LLL-A	✓			98.621 ± 0.244	97.541 ± 0.641	93.619 ± 2.726
LLL-CA	✓	✓		98.621 ± 0.244	97.634 ± 0.649	95.568 ± 0.672
LLL-AFCA	✓	✓	✓	**98.987 ± 0.359**	**97.796 ± 1.457**	**96.958 ± 0.768**

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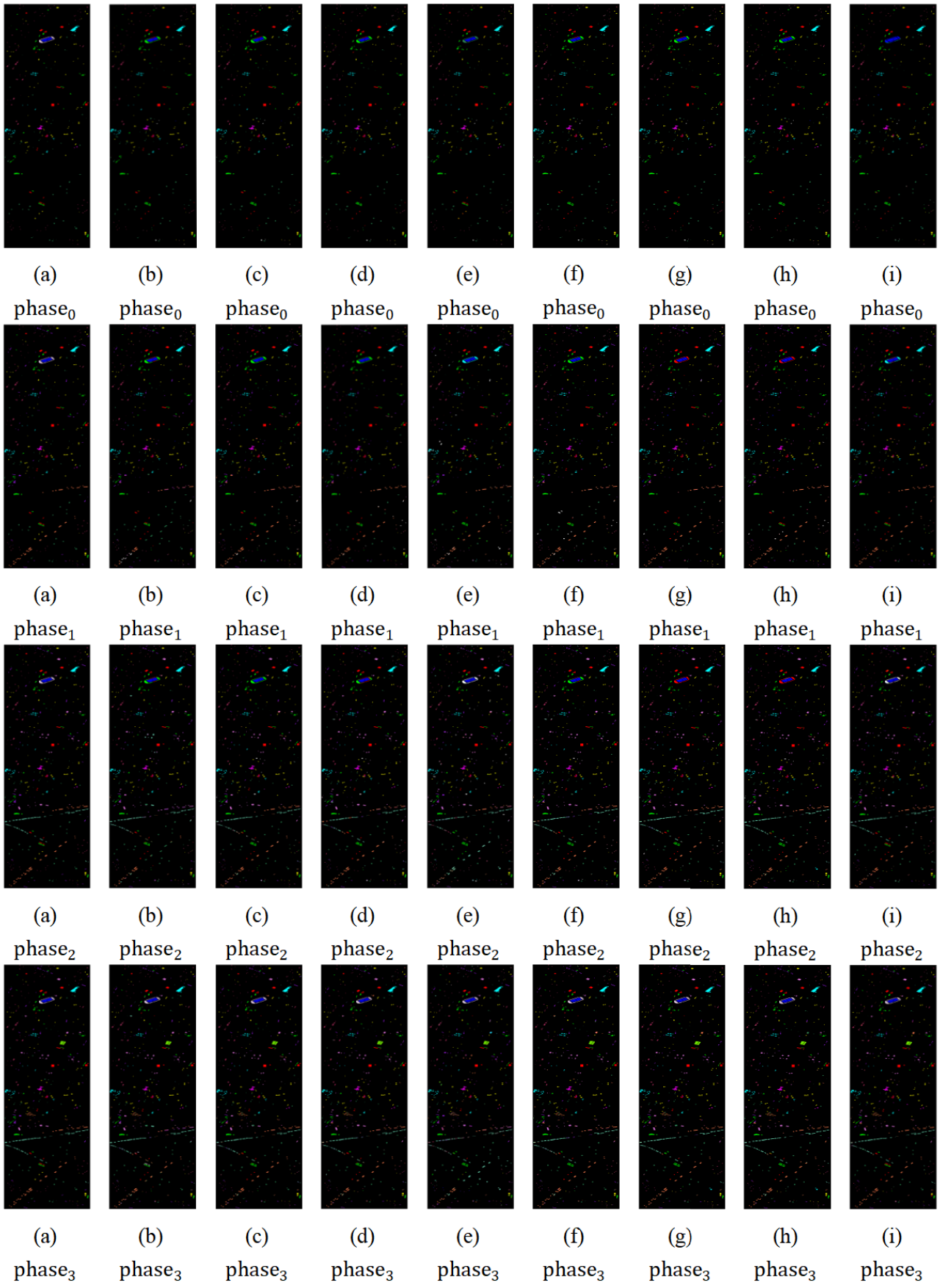


Fig. 10. Class incremental tasks classification results map of the comparative experiment on the HU dataset. (a) Ground truth. (b) LUCIR. (c) iCaRl. (d) AANet. (e) CILKD. (f) LLL-C. (g) LLL-A. (h) LLL-CA (i) LLL-AFCA.

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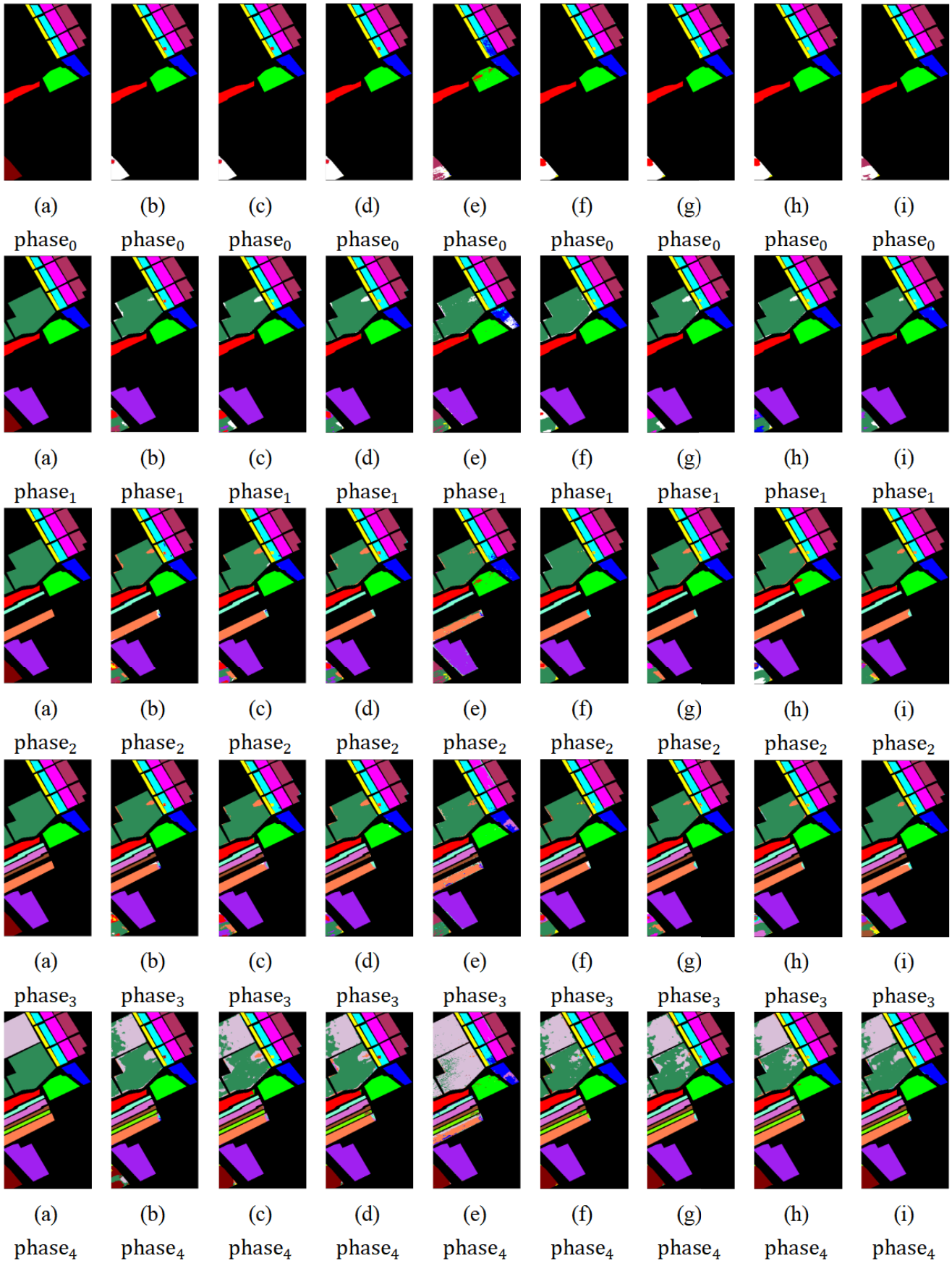


Fig. 11. Class incremental tasks classification results map of the comparative experiment on the SA dataset. (a) Ground truth. (b) LUCIR. (c) iCaRl. (d) AANet. (e) CILKD. (f) LLL-C. (g) LLL-A. (h) LLL-CA (i) LLL-AFCA.

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TABLE XVII
ABLATION EXPERIMENTS OF LLL-AFCA METHOD ON THE HU13 DATASET (%)

	A	C	M	Base phase0	Incremental phase1	Incremental phase2	Incremental phase3
Baseline				96.103 ± 0.774	79.877 ± 1.858	72.505 ± 1.445	72.163 ± 1.436
LLL-A	✓			93.131 ± 1.240	87.492 ± 2.087	85.738 ± 2.023	87.375 ± 2.081
LLL-CA	✓	✓		93.131 ± 1.240	87.860 ± 1.867	86.143 ± 2.210	87.645 ± 1.510
LLL-AFCA	✓	✓	✓	**96.425 ± 0.714**	**92.233 ± 1.519**	**90.148 ± 1.109**	**90.345 ± 1.006**

TABLE XVIII
ABLATION EXPERIMENTS OF LLL-AFCA METHOD ON THE SA DATASET (%)

	A	C	M	Base phase0	Incremental phase1	Incremental phase2	Incremental phase3	Incremental phase4
Baseline				99.583 ± 0.167	91.869 ± 1.339	88.745 ± 1.598	87.238 ± 2.261	87.238 ± 2.261
LLL-A	✓			99.407 ± 0.339	98.221 ± 0.508	98.132 ± 0.453	98.091 ± 0.535	91.363 ± 0.751
LLL-CA	✓	✓		99.407 ± 0.339	98.374 ± 0.397	98.235 ± 0.466	98.167 ± 0.555	91.535 ± 0.748
LLL-AFCA	✓	✓	✓	**99.562 ± 0.262**	**98.736 ± 0.467**	**98.482 ± 0.310**	**98.262 ± 0.368**	**92.354 ± 1.056**

TABLE XIX
ABLATION EXPERIMENTS OF LLL-AFCA METHOD ON THE HU18 DATASET (%)

Method	A	C	M	Base phase0	Incremental phase1	Incremental phase2	Incremental phase3	Incremental phase4	Incremental phase5
Baseline				75.692 ± 3.293	65.455 ± 3.497	57.355 ± 2.909	56.885 ± 2.202	50.052 ± 2.520	44.416 ± 4.102
LLL-A	✓			75.692 ± 3.293	63.087 ± 2.504	59.656 ± 2.797	59.144 ± 3.272	54.000 ± 2.758	51.359 ± 3.449
LLL-CA	✓	✓		75.692 ± 3.293	**66.234 ± 3.500**	59.671 ± 3.232	**59.824 ± 2.971**	**54.590 ± 2.813**	51.473 ± 3.269
LLL-AFCA	✓	✓	✓	**78.600 ± 2.103**	63.716 ± 2.847	**59.754 ± 2.762**	58.168 ± 2.503	53.376 ± 3.580	**53.114 ± 3.584**

phase and increased by 2.301%, 2.259%, 3.948%, and 6.943% in the second to fifth incremental phases, respectively. This indicates that adding a contrastive learning module to the shallow network can effectively extract common knowledge from both old and new classes, thereby preventing catastrophic forgetting while learning new knowledge.

After adding the contrastive learning module and class-related angular penalty loss module simultaneously, the OA for the PC dataset increased by 7.683% and 25.325% in the first and second incremental phases, respectively. For the HU dataset, the OA increased by 7.983%, 13.638%, and 15.482% in the first to third incremental phases, respectively. For the SA dataset, the OA increased by 6.505%, 9.490%, 10.929%, and 4.297% in the first to fourth incremental phases, respectively. This shows that adding an angular penalty loss module for dynamic class margin adjustment in the incremental phase can effectively reduce intraclass distance and increase interclass distance, providing more feature space for new data and reducing feature space overlap between classes, thereby improving classification accuracy. This effect is particularly significant when the number of classes is large.

Finally, after adding the data augmentation module to the model, the OA for the PC dataset increased by 7.845% and 26.715% in the first and second incremental phases, respectively. For the HU13 dataset, the OA increased by 12.356%, 12.356%, 17.643%, 17.643%, and 18.182% in the first to third incremental phases, respectively. For the SA dataset, the OA increased by 6.867%, 9.737%, 10.244%, and 5.116% in the first to fourth incremental phases, respectively. For the HU18 dataset, the OA increased by −1.739%, 2.399%, 2.869%, 1.283%, and 8.698% in the first to fifth incremental phases, respectively. This indicates that performing data augmentation in the base phase not only allows the model to classify base phase data but also enhances its ability to classify incremental phase data, improving the model's generalization and stability.

V. CONCLUSION

We propose an LLL-based HIS classification method named LLL-AFCA. Our method can achieve both retention of old knowledge and adaptation to new knowledge, avoiding catastrophic forgetting while continuously learning new classes. Specifically, we designed a dual-branch architecture where part of the network is frozen to retain old knowledge, while another part is continuously updated to learn new knowledge. To achieve adaptive fusion of old and new class knowledge, fusion weights are used after each residual block to determine the fusion proportion, and a global bilevel optimization strategy is employed to enable the model to learn the fusion weights.

On this basis, a contrastive learning module is added to the shallow network to bring the features of old and new classes closer, enabling the model to learn common features of old and new class data. In the deep network, only the knowledge adaptation block and knowledge retention block are used, allowing the model to learn the respective features of old and new class data.

In addition, an angular penalty loss for dynamic class margin adjustment is added during the incremental phase. This aims to reduce intraclass distance while increasing interclass distance, providing distinct feature space for newly learned classes and effectively reducing feature distinct overlap between old and new classes, thus improving classification accuracy.

Finally, a data augmentation strategy is used to enhance the model's generalization and stability. This allows the model to not only achieve high classification performance on current data but also to retain some classification capability for new

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data that may be encountered in future incremental phases. Extensive experiments on four datasets demonstrate the effectiveness of the LLL-AFCA method.

In the future, we will continue to explore LLL methods for HIS classification that do not require retaining old data, achieving knowledge retention using only features or networks, thereby further reducing storage and computational costs.

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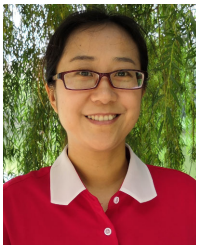
Zihui Jiang received the B.S. degree in information and computational science from the College of Science, Shenyang Aerospace University, Shenyang, China, in 2020, where she is currently pursuing the M.S. degree in computer technology with the College of Computer Science.

Her research interests include machine learning and hyperspectral image processing.



Zhaokui Li (Member, IEEE) received the M.S. degree in computer application from Liaoning University, Shenyang, China, in 2003, and the Ph.D. degree in computer software and theory from Wuhan University, Wuhan, China, in 2014.

He is currently a Professor with the School of Computer, Shenyang Aerospace University, Shenyang. His research interests include hyperspectral image analysis, computer vision, and machine learning.



Yan Wang received the B.Sc. and M.Sc. degrees from the School of Computer, Liaoning University, Shenyang, China, in 2000 and 2003, respectively.

She is currently a Lecturer with the School of Computer, Shenyang Aerospace University, Shenyang. Her research interests include image processing and pattern recognition.



Wei Li (Senior Member, IEEE) received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2007, the M.S. degree in information science and technology from Sun Yat-sen University, Guangzhou, China, in 2009, and the Ph.D. degree in electrical and computer engineering from Mississippi State University, Starkville, MS, USA, in 2012.

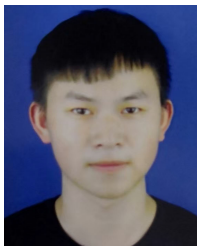
Subsequently, he spent 1 year as a Post-Doctoral Researcher with the University of California, Davis, Davis, CA, USA. He is currently a Professor with the School of Information and Electronics, Beijing Institute of Technology, Beijing, China. His research interests include hyperspectral image analysis, pattern recognition, and data compression.

Dr. Li is an Associate Editor of IEEE SIGNAL PROCESSING LETTERS (SPL) and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS).



Ke Wang received the B.S. degree in computer science and engineering from the College of Computer Science, Zaozhuang University, Shandong, China, in 2021. He is currently pursuing the M.S. degree in computer technology with the College of Computer Science, Shenyang Aerospace University, Shenyang, Liaoning, China.

His research interests include machine learning and hyperspectral image processing.



Jing Tian received the B.S. degree in computer science and technology from the College of Computer Science, Shenyang Aerospace University, Shenyang, China, in 2022, where he is currently pursuing the M.S. degree in computer science and technology.

His research interests include machine learning and hyperspectral image processing.



Chuanyun Wang received the Ph.D. degree in pattern recognition and intelligent system from Beihang University, Beijing, China, in 2017.

He is currently an Associate Professor with the School of Computer Science, Shenyang Aerospace University, Shenyang, China. His research interests include pattern recognition, swarm intelligence, and Internet of Things.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, Baltimore, MD, USA, in 2000.

She is currently the Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of the SPIE-International Society for Optics and Photonics. She received the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society. She was the Co-Chair of the Data Fusion Technical Committee of the IEEE Geoscience and Remote Sensing Society from 2009 to 2013, and the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She has served as an Associate Editor for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the *Journal of Applied Remote Sensing*, and IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She is the General Chair of the Fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing, Shanghai, in 2012.

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